

TICVAI Wallet Configuration

Backend Structure Module

10 Boards | 100 Backend Configuration Screens

Configuration Reference Guide

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Board 1 — Wallet Foundation & Wallet Types

10 Backend Configuration Screens

BOARD 1 – WALLET FOUNDATION & WALLET TYPES

Configure the core foundation of wallets including types, ownership, lifecycle, currencies, credit buckets and key features.

10 Screens Configuration Screens | 100% Configurable No Code Changes | Version Controlled Approval & Audit | Multi-Tenant Venue Aware | Role Based Access Secure & Governed

01 Wallet Command Center

Wallet Status Distribution

Category	Count	Percentage
Active	1,234,567	98.76%
Pending Activation	987,654	79.85%
Suspended	456,789	36.98%
Blocked	123,456	10.01%
Expired	567,890	45.94%
Closed	234,567	18.98%

02 Wallet Type Library

Wallet Type	Code	Category	Status	Description
Guest Wallet	GUEST	Individual	Active	Wallet for guest users
Registered Customer	CUST	Individual	Active	Wallet for registered customers
Family Wallet	FAMILY	Family	Active	Shared family wallet
Parent Wallet	PARENT	Family	Active	Parent-child wallet
Child Wallet	CHILD	Family	Active	Child's personal wallet
Corporate Wallet	CORP	Corporate	Active	Corporate account wallet
Employee Wallet	EMP	Corporate	Active	Employee benefits wallet
Membership Wallet	MEMB	Membership	Active	Membership benefits wallet
Event Wallet	EVENT	Event	Active	Event specific wallet
Rebate Wallet	REBATE	Rebate	Active	Rebate / Cashback wallet

03 Wallet Creation & Provisioning Rules

Trigger	Type	Auto Create	Status	Actions
Customer Registration	Event	No	Active	✓
Guest Checkout	Event	No	Active	✓
Membership Purchase	Event	No	Active	✓
Gift Card Activation	Event	No	Active	✓
Cardless Withdrawal	Event	No	Active	✓
Manual Administrator	Manual	No	Active	✓
API Request	API	No	Active	✓
Wallet Check-in	Event	No	Active	✓
Event Registration	Event	No	Active	✓
Corporate Account Creation	Event	No	Active	✓

04 Wallet Ownership & Account Association

Model	Description	Default Role	Status
Individual	Single user owns the wallet	Owner, User, Viewer	Active
Guest	Wallet for anonymous/guest	Owner, User	Active
Family	Shared wallet for family	Owner, Admin, Dependee	Active
Parent	Parent controls dependee	Parent, Child, Viewer	Active
Child/Dependee	Child linked to parent wallet	Child, Viewer	Active
Corporate	Organization owned wallet	Admin, User, Viewer	Active
School	School / Institution wallet	Admin, Staff, Student	Active
Employee	Employee benefits wallet	Admin, Employee	Active
Partner	Partner organization wallet	Admin, User, Viewer	Active
Tour Operator	Tour operator wallet	Admin, User	Active

05 Wallet Currency & Monetary Configuration

Base Wallet Currency: AED - UAE Dirham

Supported Currencies: AED, USD, EUR, GBP, SAR, etc.

Multi-Currency Wallet: ☒

Currency Conversion: Allowed

Exchange Rate Source: Central Bank

Exchange Rate Markup (%): 1.50

Decimal Precision: 2

Minimum Wallet Balance (AED): 0.00

Maximum Wallet Balance (AED): 1,000,000.00

Negative Balance: ☐

Zero Balance Behavior: Allow transactions

06 Credit & Balance Type Configuration

Balance Type	Code	Type	Priority	Status
Cash Credit	CASH	Monetary	1	Active
Refund Credit	REFUND	Monetary	2	Active
Bonus Credit	BONUS	Monetary	3	Active
Promotional Credit	PROMO	Monetary	4	Active
Gift Card Credit	GIFTCARD	Monetary	5	Active
Membership Credit	MEMBER	Monetary	6	Active
Ride Credit	RIDE	Non-Monetary	7	Active
Attraction Credit	ATTR	Non-Monetary	8	Active
Redemption Ticket Credit	REDEM	Non-Monetary	9	Active
Meal Credit	MEAL	Non-Monetary	10	Active

07 Wallet Feature Profile

Feature Category	Feature	Enabled
General	Customer Top-up	☒
Funding	Agent Top-up	☒
Payments	Automatic Refund	☒
Transfers	Recurring Funding	☒
Promotions	Wallet Payment	☒
Membership	Refund to Wallet	☒
Channels	Gift Cards	☒
Vouchers	Vouchers	☒
Withdrawals	ATM Transfer	☒
Security	Balance Withdrawal	☒

08 Wallet Lifecycle Configuration

Lifecycle: Draft → Pending Verification → Active → Suspended → Blocked → Expired → Closed

Status Details:

- Status Name: Active
- Description: Wallet is active and can be used for transactions
- Allows Transactions: ☒
- Allows Top-up: ☒
- Allows Transfer: ☒
- Expiry Behavior: None or Expiry after validity period
- Expiry Period (Months): 12

09 Wallet Numbering, Identity & Credentials

Wallet Number Format: (TENANT)-(VENUE)-(SEQ)

Preview: 710-ADN-000001

Profile: 710-ADN

Sequential Length: 6

Start Number: 000001

Identifier Type: ☒ QR Code, ☒ NFC, ☒ RFID

Tabulation: ☒

Credential Expiry (Months): 60

Replace Lost Credential: ☒ Allow Replacement

Allow Multiple Credentials: ☒

10 Configuration Preview, Validation & Publication

Configuration Summary:

Wallet Type	Family Wallet	Version	1.0.0
Ownership Model	Family	Status	Draft
Currency	AED	Created By	Admin User
Credit Types	9 Configured	Created On	12 May 2025 10:30
Features	26 Enabled	Last Updated	12 May 2025 10:30
Lifecycle	7 Stages	Effective Date	15 May 2025
Channels	5 Channels	Approval Required	Yes
Security Profile	Standard	Approved By	-
Financial Reporting	Configured	Published On	-
Applicable Venues	3 Venues		

Buttons: Validate, Submit for Approval, Publish

Screen 01 — Wallet Command Center

Purpose:

Provide administrators with the central operational and configuration overview of all wallet products across the TICVAI platform.

Backend Configuration:

- **Display total wallets by:**
 - Active
 - Pending Activation
 - Suspended
 - Blocked
 - Expired
 - Closed
 - Show wallet balances by tenant, venue and currency.
- **Show breakdown between:**
 - Cash value
 - Bonus value
 - Gift card value
 - Promotional credits
 - Ride credits
 - Redemption credits
 - Membership credits
 - Monitor wallet transaction volume.

- **Display today's:**
 - Top-ups
 - Spending
 - Refunds
 - Transfers
 - Adjustments
 - Highlight wallets requiring administrative attention.
 - Display wallets approaching configurable limits.
 - Show expiring credits.
 - Display abnormal balance movements.
- **Provide configurable filters for:**
 - Tenant
 - Venue
 - Wallet type
 - Currency
 - Customer type
 - Wallet status
 - Balance range
 - Date range
 - Allow authorized users to enter configuration or operational screens directly.

AI Capability:

AI highlights unusual wallet activity, unusual liability increases, rapidly changing balances and wallets requiring investigation.

Acceptance Condition:

Authorized users can obtain a consolidated wallet operational view and navigate directly to permitted wallet configuration and administrative actions.

Screen 02 — Wallet Type Library

Purpose:

Create and maintain reusable wallet types used throughout TICVAI.

Backend Configuration:

Administrators can create wallet types such as:

- Guest Wallet
- Registered Customer Wallet
- Family Wallet
- Parent Wallet
- Child Wallet
- Corporate Wallet
- School Wallet
- Employee Wallet
- Membership Wallet
- Event Wallet
- Resort Wallet
- Cashless Venue Wallet
- Closed-Loop Wallet

Each wallet type can configure:

- Wallet name
- Internal wallet code

- Description
- Wallet category
- Owning tenant
- Applicable venue
- Supported customer/account type
- Supported currencies
- Stored-value capability
- Transfer capability
- Top-up capability
- Refund capability
- Gift card support
- Voucher support
- Membership credit support
- Wearable support
- Online usage
- POS usage
- Mobile-app usage
- API usage
- Default lifecycle policy
- Default security profile
- Default financial profile

Wallet types can be:

- Draft
- Active
- Suspended
- Retired

Acceptance Condition:

Administrators can configure reusable wallet types without modifying application code.

Screen 03 — Wallet Creation & Provisioning Rules

Purpose:

Define how and when TICVAI automatically or manually creates a wallet.

Backend Configuration:

Wallet creation triggers may include:

- Customer registration
- Guest checkout
- Membership purchase
- Annual pass purchase
- Gift card activation
- Cashless wristband issuance
- Hotel/resort check-in
- Event registration
- Corporate account creation
- Family account creation
- Manual administrator creation
- API request

Configure:

- Automatic vs manual creation

- Required customer information
- Anonymous wallet permission
- Account verification requirement
- Mobile verification
- Email verification
- Digital ID verification
- Default wallet type
- Default currency
- Default status
- Initial credit
- Initial promotional value
- Wallet validity
- Wallet activation rules
- Duplicate-wallet handling
- One wallet per customer vs multiple wallets
- Maximum wallets per account

AI Capability:

AI can recommend a wallet provisioning policy based on venue type and business model.

Acceptance Condition:

The system creates wallets according to configurable business triggers, validation rules and authorization controls.

Screen 04 — Wallet Ownership & Account Association

Purpose:

Configure which customer, family or organization owns and controls a wallet.

Backend Configuration:

Supported ownership structures:

- Individual
- Guest
- Family
- Parent
- Child/dependant
- Corporate organization
- School
- Tour operator
- Partner
- Employee

Configure roles such as:

- Wallet Owner
- Wallet Administrator
- Authorized User
- Dependant
- Viewer
- Finance Controller

Permissions can determine who may:

- View balance
- Top up
- Spend

- Transfer
- Receive transfer
- View transactions
- Add payment instruments
- Add/remove wearables
- Configure spending limits
- Freeze wallet
- Receive notifications

Support:

- One wallet → multiple users
- One user → multiple wallets
- Primary wallet owner
- Secondary users
- Delegated wallet administration

This directly supports family and shared-wallet structures required by the scope.

Acceptance Condition:

Wallet ownership and delegated access can be configured independently of the wallet balance.

Screen 05 — Wallet Currency & Monetary Configuration

Purpose:

Define currencies and monetary rules applicable to wallet balances.

Backend Configuration:

Configure:

- Base wallet currency
- Permitted currencies
- Multi-currency wallet enabled/disabled
- Venue currency
- Settlement currency
- Currency conversion permitted
- Exchange-rate source
- Conversion timing
- Exchange-rate markup
- Decimal precision
- Minimum wallet balance
- Maximum wallet balance
- Negative balance permission
- Zero-balance behavior
- Currency-specific limits
- Rounding policy

Administrators can decide whether:

- Separate balance buckets exist per currency, or
- TICVAI converts wallet balances to a base wallet currency.

Acceptance Condition:

Wallet transactions consistently follow configured currency, conversion and rounding rules.

Screen 06 — Credit & Balance Type Configuration

Purpose:

Configure all value types that may be stored inside a TICVAI wallet.

The attached requirements explicitly call for multiple wallet credit types including Cash Credit, Bonus Credit, Redemption Tickets Credit and Rides Credit.

Backend Configuration:

Administrators can define:

- Cash Credit
- Refund Credit
- Bonus Credit
- Promotional Credit
- Gift Card Credit
- Membership Credit
- Loyalty Credit
- Ride Credit
- Attraction Credit
- Redemption Ticket Credit
- Meal Credit
- Retail Credit
- Parking Credit
- Event Credit
- Custom Credit Type

For each balance type configure:

- Balance code
- Display name
- Monetary/non-monetary
- Currency applicable
- Transferable
- Refundable
- Top-up allowed
- Promotional
- Expirable
- Redeemable channels
- Applicable products
- Applicable venues
- Accounting classification
- Liability classification
- Consumption priority
- Display in customer wallet

Acceptance Condition:

New wallet credit types can be introduced through configuration without development changes.

Screen 07 — Wallet Feature Profile

Purpose:

Control which capabilities are available for each wallet type.

Backend Configuration:

Feature toggles include:

- Customer top-up

- Agent top-up
- Automatic reload
- Recurring funding
- Wallet payment
- P2P transfer
- Refund-to-wallet
- Gift cards
- Vouchers
- Promotions
- Family sharing
- Corporate sharing
- Membership credits
- NFC
- RFID
- QR payment
- Wearable payment
- B2C usage
- Mobile-app usage
- POS usage
- Kiosk usage
- API usage
- Offline usage
- Balance withdrawal
- Balance transfer
- Administrative adjustment

Allow feature inheritance:

- Global → Tenant → Venue → Wallet Type → Individual Wallet

Acceptance Condition:

Features can be centrally enabled or restricted based on wallet context.

Screen 08 — Wallet Lifecycle Configuration

Purpose:

Govern the complete wallet lifecycle from creation through closure.

Backend Configuration:

Wallet statuses:

- Draft → Pending Verification → Active → Suspended → Blocked → Expired → Closed → Archived

Configure allowed transitions.

For example:

- Active → Suspended
- Active → Blocked
- Suspended → Active
- Expired → Reactivated
- Blocked → Closed

Configure:

- Activation conditions
- Verification requirements
- Suspension reasons
- Block reasons
- Expiry behavior

- Dormancy rules
- Closure requirements
- Remaining balance handling
- Reactivation rules
- Archival period
- Data retention period

Requirement 4.3.21 specifically requires the ability to block a wallet account.

Acceptance Condition:

Every status transition is controlled by permissions, configurable business rules and audit logging.

Screen 09 — Wallet Numbering, Identity & Digital Credentials

Purpose:

Configure how wallets are uniquely identified across TICVAI.

Backend Configuration:

Configure:

- Wallet ID structure
- Wallet number format
- Prefix/suffix
- Tenant code
- Venue code
- Sequential/random identifier
- Customer-facing wallet reference
- QR identifier
- NFC identifier
- RFID identifier
- Wearable token
- Mobile credential
- Digital Key / Digital ID
- Tokenization
- Credential expiry
- Credential replacement
- Lost/stolen credential handling

Support multiple credentials linked to one wallet.

For example:

- **Family Wallet #100245**

linked to:

- Father's mobile app
- Mother's mobile app
- Child wristband #01
- Child wristband #02
- RFID card
- QR credential

The source specifically calls for Digital Key authentication and linkage of venue-provided wearables with wallets.

Acceptance Condition:

Wallet credentials are uniquely generated, securely managed and independently replaceable without changing the underlying wallet.

Screen 10 — Wallet Configuration Preview, Validation & Publication

Purpose:

Provide a governed final validation step before wallet configuration becomes operational.

Backend Configuration:

Show configuration summary covering:

- Wallet type
- Ownership model
- Currency
- Credit types
- Features
- Lifecycle
- Credentials
- Financial configuration
- Security profile
- Applicable venues
- Applicable channels
- Effective date

Automatically validate:

- Conflicting settings
- Missing currency
- Missing financial mapping
- Unsupported credit types
- Invalid lifecycle transitions
- Missing permissions
- Missing security policy
- Unsupported channel combinations
- Wallet balance-rule conflicts
- Duplicate wallet codes

Provide:

- Draft
- Validate
- Submit for Approval
- Approve
- Reject
- Schedule Publication
- Publish
- Clone Configuration
- Create New Version
- **Compare Versions**

Every configuration publication records:

- User
- Date/time
- Old value
- New value
- Reason
- Approval
- Effective date

AI Capability:

AI performs a configuration health check and explains potential risks before publication.

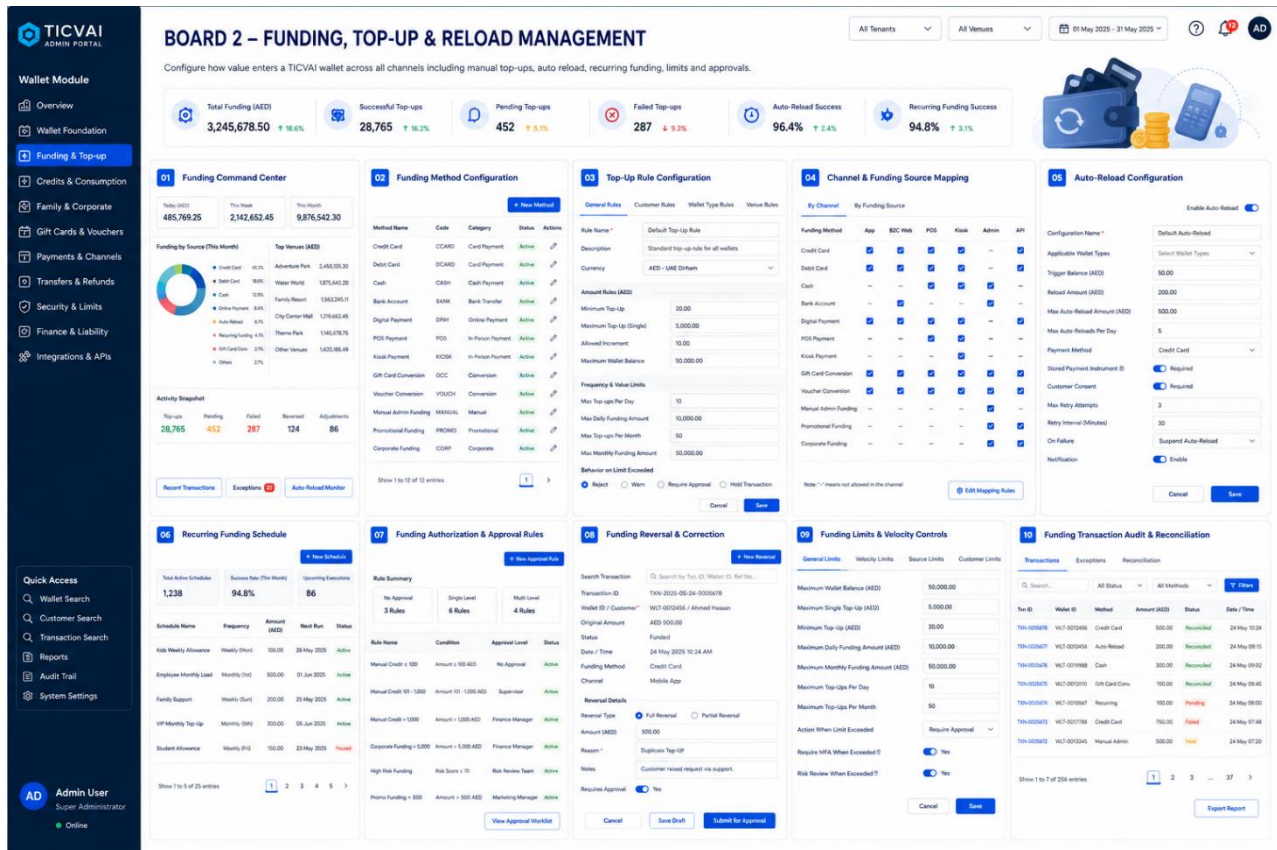
Acceptance Condition:

No wallet configuration can become operational unless validation, permissions and applicable approval requirements have been successfully completed.

This makes **Board 1 the foundation of the entire wallet engine**, rather than jumping immediately into transactions. Once these 10 screens exist, Boards 2–10 can reuse the Wallet Type, Credit Type, Currency, Ownership and Lifecycle objects without recreating them.

The next board should therefore be **Board 2 — Funding, Top-Up & Reload Management**, covering cash/card top-up, payment-source rules, auto-reload, recurring funding, funding authorization, reversals and funding limits.

Board 2 — Funding, Top-Up & Reload Management



10 Backend Configuration Screens

Board Objective: Configure how value enters a TICVAI wallet across digital and physical channels, including manual top-ups, payment-method funding, automatic reload, recurring funding, authorization controls, reversals, limits, and operational monitoring.

This board directly covers the wallet requirements for multiple funding methods, auto-reload, recurring funding, configurable limits, security controls, and complete transaction auditing.

Screen 01 — Funding Command Center

Purpose:

Provide administrators and finance/operations teams with a centralized view of all wallet funding activity.

Backend Configuration:

- **Display total funding by:**
 - Today
 - Week
 - Month
 - Custom period
- **Show funding by source:**
 - Credit card
 - Debit card
 - Cash
 - Online payment
 - Gift card conversion
 - Bank-linked funding
 - Kiosk

- POS
- Manual adjustment
- Auto-reload
- Recurring funding
- **Show:**
 - Successful top-ups
 - Pending top-ups
 - Failed top-ups
 - Reversed top-ups
 - Declined transactions
 - Monitor funding by tenant, venue and currency.
 - Display average top-up value.
 - Highlight unusual funding activity.
 - Show auto-reload performance.
 - Show recurring funding success/failure.
- **Filter by:**
 - Wallet type
 - Customer
 - Funding method
 - Channel
 - Venue
 - Currency
 - Transaction status
 - Date
 - Drill down to individual funding transactions.

AI Capability:

AI identifies abnormal funding patterns, unusual increases in wallet funding, repeated failed top-ups and potential misuse.

Acceptance Condition:

Authorized users can monitor wallet funding activity across all configured channels and investigate individual transactions with complete traceability.

Screen 02 — Funding Method Configuration

Purpose:

Configure which funding methods TICVAI wallets can accept.

The source requires multiple methods for funding wallet value, including bank-linked funding, credit/debit cards through digital channels, and cash/cards through physical channels.

Backend Configuration:**Create funding methods such as:**

- Credit Card
- Debit Card
- Cash
- Bank Account
- Digital Payment
- POS Payment
- Kiosk Payment
- Gift Card Conversion
- Voucher Conversion
- Manual Administrative Funding

- Promotional Funding
- Corporate Funding

For each method configure:

- Funding method name
- Internal code
- Funding category
- Payment service/provider
- Supported currencies
- Minimum top-up
- Maximum top-up
- Applicable wallet types
- Applicable credit types
- Applicable tenants
- Applicable venues
- Applicable channels
- Customer verification requirement
- Authorization requirement
- Refund/reversal eligibility
- Fees
- Effective dates
- Active/inactive status

Acceptance Condition:

Administrators can introduce, modify or deactivate wallet funding methods through configuration without changing application code.

Screen 03 — Top-Up Rule Configuration

Purpose:

Control the business rules governing individual wallet top-up transactions.

Backend Configuration:

Configure:

- Minimum top-up amount
- Maximum top-up amount
- Allowed increments
- Maximum number of top-ups per day
- Maximum number per week/month
- Maximum daily funding value
- Maximum monthly funding value
- Wallet maximum-balance check
- Customer-specific limits
- Wallet-type-specific limits
- Currency-specific limits
- Channel-specific limits
- Venue-specific limits
- Funding-source restrictions
- Anonymous-wallet restrictions
- Membership-specific rules
- Corporate-wallet rules
- Promotional top-up rules

Define behavior when limits are exceeded:

- Reject
- Warn
- Request approval
- Hold transaction
- Route for risk review

Support hierarchy:

- Global → Tenant → Venue → Wallet Type → Customer/Wallet

Acceptance Condition:

Every wallet top-up is evaluated against the applicable funding rules before value is credited.

Screen 04 — Channel & Funding Source Mapping

Purpose:

Determine where each funding method may be used.

Backend Configuration:**Configure funding availability across:**

- B2C Website
- Customer Mobile App
- POS
- Mobile POS
- Kiosk
- Guest Services
- Call Center
- Back Office
- Corporate Portal
- Partner/Reseller
- API
- Third-party systems

Example:

Funding Source	App	B2C	POS	Kiosk	Admin	API
Credit Card	✓	✓	✓	✓	—	✓
Cash	—	—	✓	✓	✓	—
Gift Card	✓	✓	✓	✓	✓	✓
Manual Credit	—	—	—	—	✓	—

Configure:

- Channel eligibility
- Wallet type
- Credit type generated
- Funding limits
- Authentication requirements
- Applicable venue
- Applicable customer type
- Effective dates

Acceptance Condition:

TICVAI permits wallet funding only through explicitly configured and authorized channel/source combinations.

Screen 05 — Auto-Reload Configuration

Purpose:

Configure automatic wallet replenishment when balances fall below predefined thresholds.

The attached scope specifically requires automatic top-up when a wallet balance falls below a configurable threshold.

Backend Configuration:

Configure:

- **Trigger Balance**

Example:

- When balance falls below AED 50
- **Reload Amount**
- Automatically add AED 200

Configure:

- Eligible wallet types
- Eligible credit types
- Trigger balance
- Reload amount
- Maximum reload amount
- Maximum auto-reloads per day
- Maximum auto-reloads per month
- Payment method
- Stored payment instrument
- Customer consent required
- Consent validity
- Failed payment behavior
- Retry rules
- Maximum retries
- Notification rules
- Suspension after repeated failure
- Funding limits
- Effective dates

Allow customer-level override where permitted.

Acceptance Condition:

Auto-reload occurs only after the configured threshold is reached and all consent, payment, security and funding-limit validations have passed.

Screen 06 — Recurring Funding Schedule

Purpose:

Configure scheduled wallet funding independent of the wallet's current balance.

The source specifically requires scheduled or recurring top-ups based on date, frequency, amount, payment method and customer authorization.

Backend Configuration:

Support schedules such as:

- Daily
- Weekly
- Monthly
- Specific day of month
- Custom recurrence

Examples:

Parent transfers AED 100 every Monday to Child Wallet.

Corporate wallet allocates AED 500 monthly to each authorized employee.

Customer automatically adds AED 300 on the first day of every month.

Configure:

- Schedule name
- Source wallet/payment method
- Destination wallet
- Amount
- Credit type
- Start date
- End date
- Frequency
- Execution time
- Maximum occurrences
- Funding limits
- Customer authorization
- Retry policy
- Failure behavior
- Notification rules
- Pause/resume
- Cancellation

Acceptance Condition:

TICVAI executes recurring wallet funding according to the approved schedule while enforcing applicable limits and authorization requirements.

Screen 07 — Funding Authorization & Approval Rules

Purpose:

Apply governance and approval controls to wallet funding operations.

Backend Configuration:**Configure approval based on:**

- Funding amount
- Funding method
- Wallet type
- Customer type
- Channel
- Credit type
- Administrative user
- Venue
- Tenant
- Risk score

Example:

- **Manual credit ≤ AED 100**
→ No approval
- **AED 101–1,000**
→ Supervisor approval
- **Above AED 1,000**
→ Finance Manager approval

Configure:

- Single approval
- Multi-level approval
- Maker-checker control
- Role-based approval
- Amount threshold
- Approval SLA
- Escalation
- Auto-rejection after expiry
- Mandatory reason
- Supporting documentation
- Approval notifications

High-risk activity can be routed to the Wallet Security controls covered later in Board 8.

Acceptance Condition:

Funding transactions requiring approval cannot alter the wallet balance until all configured authorization requirements are satisfied.

Screen 08 — Funding Reversal & Correction Management

Purpose:

Provide controlled correction of incorrectly funded wallet transactions without deleting financial history.

Backend Configuration:**Support:**

- Full reversal
- Partial reversal
- Duplicate top-up correction
- Failed funding correction
- Administrative correction
- Payment reversal
- Chargeback-related correction
- Wrong-wallet correction

Configure:

- Eligible transaction status
- Maximum reversal period
- Full/partial reversal permission
- Required reason code
- Required notes
- Approval threshold
- Maker-checker requirement
- Original funding reference
- Related payment reference
- Financial posting requirement
- Customer notification
- Credit expiry impact

Original transactions must remain immutable.

Instead of deleting:

- **Original Top-Up + AED 500**

TICVAI records:

- **Reversal – AED 500**

Both transactions remain linked.

Acceptance Condition:

Every funding correction creates a traceable compensating transaction while preserving the original transaction and complete audit history.

Screen 09 — Funding Limits & Velocity Controls

Purpose:

Protect wallets against excessive or suspicious funding activity.

The source specifically requires configurable maximum balance, daily top-up limit, single-transaction limit and user-specific limits.

Backend Configuration:

Configure:

- **Balance Limits**
 - Maximum wallet balance
 - Maximum credit-type balance
- **Transaction Limits**
 - Minimum funding
 - Maximum single top-up
- **Velocity Limits**
 - Maximum top-ups per hour
 - Maximum top-ups per day
 - Maximum daily amount
 - Maximum weekly amount
 - Maximum monthly amount
- **Source Limits**
 - Maximum per payment instrument
 - Maximum cash funding
 - Maximum gift-card conversion
 - Maximum promotional funding
- **Customer Limits**
 - Individual
 - Child
 - Family
 - Corporate
 - Employee
 - Guest

Actions when a rule is triggered:

- Allow
- Warn
- Decline
- Hold
- Require MFA
- Require approval
- Send to risk review
- Temporarily disable funding

AI Capability:

AI evaluates funding velocity alongside historical wallet behavior and identifies activity that is unusual even when individual transactions remain within configured limits.

Acceptance Condition:

Funding limits are evaluated in real time before wallet value is credited, with configurable actions for violations.

Screen 10 — Funding Transaction Audit & Reconciliation

Purpose:

Provide a complete administrative record of every value entering a wallet.

The source requires a complete audit trail for wallet top-ups, payments, refunds, transfers, adjustments, expirations, reversals and administrative actions.

Backend Configuration & View:

Each funding record contains:

- Wallet transaction ID
- Wallet ID
- Customer/account
- Funding source
- Payment transaction reference
- Funding method
- Channel
- Venue
- Tenant
- Currency
- Gross amount
- Fee
- Net wallet credit
- Credit type
- Date/time
- User/device
- Approval reference
- Transaction status
- Reversal reference
- Risk status
- External provider reference

Transaction statuses:

- **Initiated → Authorized → Funded → Reconciled**

Exception statuses:

- **Pending / Failed / Declined / Held / Reversed / Partially Reversed**

Provide reconciliation status between:

- **Payment Transaction ↔ Wallet Ledger Transaction**

Allow administrators to identify:

- Payment succeeded but wallet not funded
- Wallet funded but payment unresolved
- Duplicate funding
- Missing external reference
- Amount mismatch
- Currency mismatch
- Unreconciled transaction

Provide export and reporting for authorized finance users.

AI Capability:

AI identifies reconciliation exceptions and prioritizes discrepancies based on financial value, age and risk.

Acceptance Condition:

Every wallet funding transaction is traceable from its originating funding/payment event through the wallet ledger, with discrepancies clearly identified and no financial transaction removable from audit history.

-
- Board 2 Flow

The backend flow is therefore:**Funding Method**

- Channel Eligibility
- Top-Up Rules
- Limits & Risk Checks
- Authorization/Approval
- Payment/Funding Execution
- Wallet Ledger Credit
- Audit & Reconciliation

This also keeps the architecture clean: **Board 2 controls wallet funding**, but the actual card acquiring, EMV terminal, payment gateway and banking functions remain responsibilities of the Payment module rather than being duplicated inside Wallet.

Board 3 — Stored Value, Credits & Consumption Rules

10 Backend Configuration Screens

BOARD 3 – STORED VALUE, CREDITS & CONSUMPTION RULES

Configure wallet credit types, usage, expiry, priority, FEFO consumption and multi-credit behavior across all TICVAI services.

Summary Metrics:

- Total Wallet Liability: AED 18,765,432.50
- Outstanding Balance: AED 14,523,210.30
- Redeemed (This Month): AED 2,854,210.80
- Expired (This Month): AED 325,410.20
- Credits Issued (This Month): AED 3,165,432.10
- Credits Expiring Soon (30 Days): AED 1,245,682.90

01 Stored Value & Credit Command Center

Liability by Credit Type (AED)

Credit Type	Balance	Percentage
Cash Credit	4,345,210.50	29.3%
Bonus Credit	2,345,678.90	16.1%
Refund Credit	1,234,567.89	8.5%
Gift Card Credit	2,109,876.54	14.5%
Promotional Credit	1,234,567.89	8.5%
Membership Credit	1,234,567.89	8.5%
Ride Credit	1,234,567.89	8.5%
Attraction Credit	1,234,567.89	8.5%
Redemption Ticket Credit	1,234,567.89	8.5%
F&B / Meal Credit	1,234,567.89	8.5%
Retail Credit	1,234,567.89	8.5%

02 Credit Type Definition Studio

Credit Type: Code, Category, Issuable, Available, Status, Active

Credit Type	Code	Category	Issuable	Available	Status	Active
Cash Credit	CASH	Stored Value	Yes	Yes	Active	✓
Bonus Credit	BONUS	Promotional	Yes	Yes	Active	✓
Refund Credit	REFUND	Adjustment	Yes	Yes	Active	✓
Gift Card Credit	GIFT	Gift	Yes	Yes	Active	✓
Membership Credit	MEMBER	Membership	Yes	Yes	Active	✓
Ride Credit	RIDE	Attraction	Yes	Yes	Active	✓
Attraction Credit	ATTRA	Attraction	Yes	Yes	Active	✓
F&B / Meal Credit	F&B	F&B	Yes	Yes	Active	✓
Retail Credit	RETAIL	Retail	Yes	Yes	Active	✓
Parking Credit	PARK	Parking	Yes	Yes	Active	✓

03 Credit Issuance Rule Configuration

Insurance Sources, Rules, Mappings, Approvals

04 Credit Usage & Eligibility Rules

Credit Type: Meal Credit

Rule Name: Meal Credit - F&B Outlets

Description: Application only in F&B outlets during operating hours

Eligible Products: F&B Outlets

Eligible Categories: Food, Non-Alcoholic Beverages

Excluded Categories: Alcoholic, Tobacco

Allowed Venues: Waterpark, Theme Park

Allowed Channels: POS, In-Store App, Self-Service

Time Window: 11:00 AM - 10:00 PM

Days of Week: Daily

Minimum Spend: AED 0.00

Stackable with other credits: ☒

05 Consumption Priority Engine

Policy Name: Default Consumption Priority

Applies To: All Wallet Types

Applies To Channel: All Channels

Applies To Venues: All Venues

06 Expiry & Validity Policy Configuration

Policy: Notifications, Extensions

Credit Type: Bonus Credit

Validity Model: Fixed validity period from issuance

Validity Period: 90 Days

Grace Period: 7 Days

Unlimited Validity: ☒

Extension Allowed: ☒

Max Extension Period: 30 Days

Renewal On: Membership Renewal

Action On Expiry: Forfeit Credit

Accounting Treatment: Mark as Breakage / Expired Liability

07 FEFO & Credit Lot Management

Lot Selection, Settings, Behavior

Credit Type: Bonus Credit

Lot Consumption Strategy: FEFO - First Expiry, First Out

Chop Lot Consumption: ☒

Allow Period Lot Consumption: ☒

Lock Lot During Reservation: ☒

Reserve Lot On Reservation: ☒

Handle Expired Lot: Move to Expired Bucket

Display Lot Details To Agents: ☒

08 Split Tender & Multi-Credit Consumption

General, Insufficient Balance Behavior, Rounding

09 Expiry, Extension & Forfeiture Operations

Expiring Today: 1,245

Expiring 7 Days: 4,532

Expired: 2,145

Overdue (This Month): 186

10 Consumption Simulator, Validation & Publication

Simulator, Validation, Versions

Customer / Wallet: John Doe - Wallet #123456789

Venue: Waterpark

Channel: POS

Product / Service: Family Meal

Transaction Amount (AED): 250.00

Transaction Date: 28 Aug 2025 03:30 PM

Simulation Result:

Item	Amount (AED)	Status
Meal Credit	60.00	✓ Eligibility Rules
Bonus Credit	50.00	✓ Validity / Expiry
Gift Card Credit	80.00	✓ Consumption Priority
Cash Credit	60.00	✓ FEFO Selection
Total Allowed (AED)	250.00	
Remaining Amount To Pay	0.00	✓ Limits Check

Board Objective: Configure the intelligence behind the TICVAI wallet balance: what types of value can be stored, where each credit can be spent, when it expires, which balance is consumed first, how FEFO operates, and how credits behave across Ticketing, F&B, Retail, Attractions, Parking, Membership and other TICVAI services.

This board directly addresses requirements including multiple credit types, configurable expiry, usage criteria, FEFO consumption, stored-value balances, membership-related credits, and wallet reporting.

Screen 01 — Stored Value & Credit Command Center

Purpose:

Provide administrators with a centralized operational overview of all stored value and digital credits held across TICVAI wallets.

Backend Configuration & Monitoring:

Display total outstanding value by:

- Cash Credit
- Bonus Credit
- Refund Credit
- Gift Card Credit
- Promotional Credit
- Membership Credit
- Ride Credit
- Attraction Credit
- Redemption Ticket Credit
- F&B / Meal Credit
- Retail Credit

- Parking Credit
- Custom Credit

Show:

- Total issued value
- Current outstanding balance
- Redeemed value
- Expired value
- Value expiring soon
- Suspended/frozen value
- Average wallet balance
- Credits issued today
- Credits consumed today

Filter by:

- Tenant
- Venue
- Wallet type
- Credit type
- Customer/account type
- Currency
- Source
- Expiry period
- **Status**

Provide drill-down from aggregated liability into individual wallets and credit transactions.

AI Capability:

AI highlights abnormal credit issuance, unusually high outstanding balances, rapidly increasing liabilities and credit types with unusual redemption or expiry patterns.

Acceptance Condition:

Authorized administrators can monitor all wallet credit balances and trace aggregate figures to their originating wallet and ledger transactions.

Screen 02 — Credit Type Definition Studio

Purpose:

Create and maintain the individual value buckets that can exist inside a TICVAI wallet.

The source explicitly requires support for different digital wallet credit types such as **Cash Credit, Bonus Credit, Redemption Tickets Credit and Rides Credit**, with configurable usage criteria.

Backend Configuration:**For each credit type configure:**

- Credit name
- Internal code
- Description
- Credit category
- Monetary / non-monetary
- Currency
- Unit of measure
- Decimal precision
- Transferable
- Refundable
- Reversible

- Top-up eligible
- Customer purchasable
- Promotional
- Membership related
- Gift related
- Expirable
- Shareable
- Family eligible
- Corporate eligible
- Offline usage permitted
- Customer-visible
- Accounting classification
- Liability classification

Example credit types:

- **CASH**
Cash Credit
- **BONUS**
Bonus Credit
- **RIDE**
Ride Credit
- **MEAL**
Meal Credit
- **MEMBER**
Membership Credit
- **PARK**
Parking Credit
- **REDEEM**
Redemption Ticket Credit

Acceptance Condition:

Administrators can introduce and configure new stored-value or entitlement credit types without software development.

Screen 03 — Credit Issuance Rule Configuration

Purpose:

Define how credits enter a wallet after the wallet itself has already been funded or as the result of another TICVAI business process.

This screen is deliberately different from Board 2: **Board 2 controls funding/payment into a wallet; this screen controls the creation of individual credit buckets and entitlements.**

Issuance Sources:

- Wallet top-up
- Ticket purchase
- Membership purchase
- Membership renewal
- Promotion
- Loyalty reward
- Gift card activation
- Voucher conversion
- Refund
- Customer service compensation
- Event campaign
- Corporate allocation

- Family allocation
- Manual administrator action
- API/integration
- Business rule
- AI-driven campaign

Configure:

- Source event
- Credit type
- Fixed amount
- Percentage-based amount
- Quantity
- Maximum issuance
- Customer eligibility
- Applicable wallet
- Applicable venue
- Effective dates
- Approval requirement
- Expiry policy
- Accounting treatment

Example:

- **Gold Membership Purchase**

→ Issue 5 Guest Credits

→ Issue AED 100 F&B Credit

→ Issue 2 Parking Credits.

Acceptance Condition:

Configured business events generate the correct wallet credits with appropriate source references, expiry rules and audit history.

Screen 04 — Credit Usage & Eligibility Rules

Purpose:

Define exactly **where** and **under what conditions** each wallet credit may be consumed.

Backend Configuration:

Credits may be restricted by:

- **Product**
- Ticket type
- Membership
- Attraction
- Ride
- F&B item/category
- Retail item/category
- Rental product
- Parking
- Event
- Add-on
- **Location**
- Tenant
- Venue
- Attraction
- Store

- Restaurant
- POS location
- **Channel**
- POS
- B2C
- Mobile App
- Kiosk
- API
- Wearable
- **Customer**
- Membership tier
- Customer segment
- Age
- Family role
- Corporate role
- **Time**
- Date
- Day of week
- Time of day
- Season
- Event period

Example:

- **Meal Credit**

Valid only at F&B outlets

Venue: Waterpark

11:00–18:00

Excludes alcohol/specified categories

Expires same day.

Acceptance Condition:

The wallet engine permits redemption only when the transaction satisfies all applicable credit eligibility rules.

Screen 05 — Consumption Priority Engine

Purpose:

Configure which wallet balance TICVAI should consume first when multiple eligible credits can pay for the same transaction.

This directly addresses the source requirement that usage priority must be configurable—for example, consuming Cash Credit before Bonus Credit.

Backend Configuration:**Administrators define priority sequences such as:**

- Expiring Promotional Credit
- Bonus Credit
- Gift Card Credit
- Membership Credit
- Cash Credit
- External Payment

Or:

- Membership Included Credit
- Ride Credit
- Bonus Credit

- Cash Credit

Configure priority by:

- Tenant
- Venue
- Wallet type
- Product
- Channel
- Customer type
- Membership
- Credit type

Support:

- Fixed priority
- Customer-selectable priority
- System-optimized priority
- FEFO priority
- Restricted credits first
- Promotional credits first
- Cash value last

Simulation Tool:

Administrator enters:

- Transaction: AED 150 F&B

TICVAI displays:

- Meal Credit: AED 40
Bonus Credit: AED 30
Cash Credit: AED 80

and explains why each balance was selected.

Acceptance Condition:

When multiple credits are eligible, TICVAI deterministically selects balances according to the published consumption policy.

Screen 06 — Expiry & Validity Policy Configuration

Purpose:

Configure how long each type of wallet value remains valid.

The source requires configurable expiry periods for all credit types, including the ability to configure **unlimited validity**, particularly for cash credit.

Expiry Models:

- Never expires
- Fixed expiry date
- X days from issuance
- X days from activation
- End of month
- End of season
- Membership expiry
- Event expiry
- Ticket validity expiry
- Custom rule

Configure:

- Valid-from date

- Expiry calculation
- Grace period
- Expiry timezone
- Extension allowed
- Manual extension permission
- Maximum extension
- Renewal behavior
- Remaining-value behavior
- Notification schedule
- Accounting treatment after expiry

Example:

- **Cash Credit**
→ Unlimited
- **Bonus Credit**
→ 90 days
- **Event Meal Credit**
→ End of event day
- **Membership Credit**
→ Membership expiry

Acceptance Condition:

Each issued credit receives the correct validity period based on the applicable published expiry policy.

Screen 07 — FEFO & Credit Lot Management

Purpose:

Control consumption when multiple lots of the **same credit type** have different expiry dates.

Requirement 4.3.19 specifically requires TICVAI to support **FEFO — First Expiry, First Out**.

For example, the customer has:

Credit Lot	Balance	Expiry
Bonus A	AED 50	28 Aug
Bonus B	AED 100	15 Sep
Bonus C	AED 200	31 Dec

For an AED 120 eligible transaction:

TICVAI consumes:

- **AED 50 — Bonus A**
- then
- **AED 70 — Bonus B**

leaving:

- **AED 30 — Bonus B**
AED 200 — Bonus C

Backend Configuration:

Support strategies:

- FEFO
- FIFO
- Oldest issuance first
- Highest restriction first
- Custom priority

Configure:

- Strategy per credit type
- Cross-lot consumption
- Partial lot consumption
- Lot locking
- Reserved balance handling
- Expired lot handling
- Reversal behavior
- Refund restoration behavior

Acceptance Condition:

TICVAI tracks credit lots independently and consumes them according to the configured lot-selection strategy.

Screen 08 — Split Tender & Multi-Credit Consumption

Purpose:

Configure transactions where multiple wallet credits and external payment methods are combined.

Example:

- **Transaction**
- **AED 300**

Available:

- Meal Credit: AED 50
- Bonus Credit: AED 40
- Gift Card: AED 100
- Cash Wallet: AED 70

Wallet covers:

- **AED 260**

Remaining:

- **AED 40**

Customer pays AED 40 by card.

Backend Configuration:**Configure:**

- Multi-credit usage allowed
- Maximum credit types per transaction
- Partial redemption
- Wallet + card
- Wallet + cash
- Wallet + voucher
- Wallet + loyalty
- Wallet + gift card
- Wallet + multiple payment methods
- Minimum external payment
- Rounding
- Insufficient-wallet behavior

Options when balance is insufficient:

- Request external payment
- Reject transaction
- Use next eligible wallet
- Ask customer

- Allow authorized negative balance
- Route to corporate wallet

This also supports the attached requirement where a wallet-linked wristband may have insufficient value and the outstanding amount must be settled.

Acceptance Condition:

TICVAI correctly allocates transaction value across eligible credits and external tender while maintaining a complete allocation record.

Screen 09 — Credit Expiry, Extension & Forfeiture Operations

Purpose:

Provide governed administrative management of credits approaching or reaching expiry.

Backend Operations:

Display:

- Expiring today
- Expiring in 7 days
- Expiring in 30 days
- Expired
- Extended
- Forfeited
- Suspended

Authorized administrators may:

- Extend expiry
- Restore expired credit
- Suspend credit
- Reinstate credit
- Forfeit credit
- Transfer eligible credit
- Correct expiry
- Bulk extend credits

Require:

- Reason code
- User notes
- Approval where applicable
- Effective date
- Supporting reference

Configure customer notifications:

- 30 days before expiry
- 14 days
- 7 days
- 1 day
- On expiry

The source explicitly requires guests to be notified about expiring wallet credits.

AI Capability:

AI can identify significant balances approaching expiry and recommend targeted customer communications or campaigns, subject to configured business rules.

Acceptance Condition:

Expiry-related administrative actions preserve the original credit history and create immutable audit entries.

Screen 10 — Consumption Simulator, Validation & Rule Publication

Purpose:

Allow administrators to test the entire credit engine before publishing new rules.

This is important because the interaction between eligibility, expiry, FEFO and priority can otherwise create unintended wallet behavior.

- **Test Scenario**

Administrator selects:

Customer: Gold Member

Wallet: Family Wallet

Venue: Theme Park

Product: Family Meal

Transaction: AED 250

Channel: POS

Date: 28 August

TICVAI evaluates:

- **Available Credits**
- Meal Credit — AED 60
- Bonus Credit — AED 50
- Gift Card — AED 80
- Cash Credit — AED 200

Then displays:

- **Eligibility → PASS**
- **Expiry Evaluation → PASS**
- **FEFO Selection → Bonus Lot #03**
- **Consumption Result**
- Meal Credit → AED 60
- Bonus Credit → AED 50
- Gift Card → AED 80
- Cash Credit → AED 60
- **Total = AED 250**

The simulator should explain:

- Which rules fired
- Which credits qualified
- Which credits were rejected
- Why a credit was rejected
- Consumption sequence
- Lot selection
- Expiry impact
- Remaining balances

Publication controls:

- Save Draft
- Validate
- Compare Version
- Submit for Approval
- Approve
- Schedule
- Publish
- Roll Back

AI Capability:

AI identifies conflicting consumption rules, unreachable credit types, incorrect priorities and potentially unintended financial outcomes before publication.

Acceptance Condition:

No consumption-rule configuration becomes active until it passes validation and the required approval/publication workflow.

Board 3 — Core Engine Flow

The backend logic should effectively operate as:

Transaction Request

- Identify Available Credit Types
- Check Product/Channel/Venue Eligibility
- Check Validity & Expiry
- Apply Consumption Priority
- Apply FEFO Within Credit Lots
- Calculate Split Tender
- Reserve Value
- Consume Value
- Update Wallet Ledger & Remaining Balances

The distinction between the first three boards is therefore very clean:

Board 1 — What is the wallet?

Wallet structure, type, ownership, currency and lifecycle.

Board 2 — How does value enter the wallet?

Funding, top-ups, reloads, limits and funding reconciliation.

Board 3 — How is stored value created, governed and consumed?

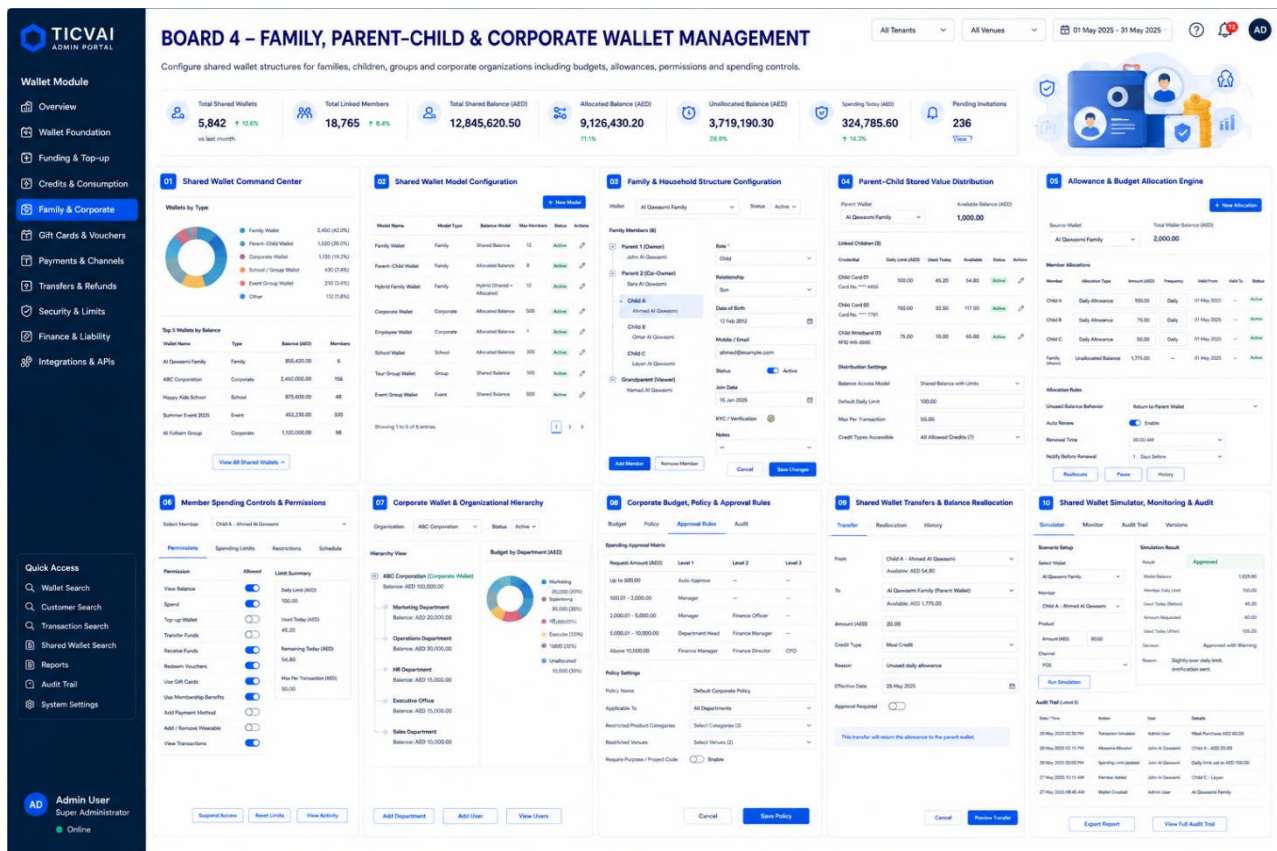
Credits, eligibility, priority, expiry, FEFO and split consumption.

This avoids duplication and gives a clear backend domain model to implement.

Board 4 should be Family, Parent–Child & Corporate Wallet Management, where we configure shared balances, parent/child relationships, allowances, spending caps, delegated authority, corporate budgets and the amusement-park parent-card/child-card requirement from **4.3.11, 4.3.26, 4.3.27 and 4.3.38**.

Board 4 — Family, Parent–Child & Corporate Wallet Management

10 Backend Configuration Screens



Board Objective: Configure TICVAI's shared-wallet architecture for families, parents and children, groups, schools, companies, corporate clients and other organizations.

This board covers the requirements for **family wallets, designated wallet owners, budget distribution, spending caps, shared balances, corporate spending wallets, permissions, and parent-card/child-card stored-value distribution.**

The central concept is that TICVAI should support both:

Shared Balance Model — multiple authorized users consume one central balance.

- and

Allocated Balance Model — the wallet owner distributes allowances/budgets to linked users while retaining centralized control.

Screen 01 — Shared Wallet Command Center

Purpose: Provide administrators with a centralized operational view of all Family, Parent-Child and Corporate wallets.

- **Backend Configuration & Monitoring**

Display:

- Total shared wallets
- Family wallets
- Parent-child wallets
- Corporate wallets
- School/group wallets
- Active linked users
- Total shared balance
- Allocated balance

- Unallocated balance
- Spending today
- Transfers today
- Wallets approaching limits
- Suspended child/member access
- Pending invitations/associations

Break down wallets by:

- Tenant
- Venue
- Wallet model
- Customer type
- Currency
- **Status**

Provide quick actions:

- Create Shared Wallet
- Add Member
- Allocate Budget
- Adjust Limits
- Suspend Member
- View Transactions
- Investigate Exceptions

AI Capability:

AI highlights unusual dependent spending, abnormal corporate expenditure, inactive allocations, unusual transfers and wallets likely to exceed configured budgets.

Acceptance Condition:

Authorized administrators can monitor shared-wallet structures and drill down from consolidated information to individual wallets, members, allocations and transactions.

Screen 02 — Shared Wallet Model Configuration

Purpose: Define reusable shared-wallet structures.

Administrators can create models such as:

- Family Wallet
- Parent–Child Wallet
- Corporate Wallet
- Employee Wallet
- School Wallet
- Student Wallet
- Tour Group Wallet
- Event Group Wallet
- Hospitality Group Wallet
- **Balance Models**
- **Model A — Fully Shared Balance**
- Parent:
AED 1,000

Children consume directly from the same AED 1,000 balance.

- **Model B — Allocated Balance**
- Parent:
AED 1,000

- → Child A: AED 200
→ Child B: AED 150
→ Parent Available: AED 650
- **Model C — Shared Balance + Individual Limits**

All users consume the central balance, but:

Child A → Max AED 100/day

Child B → Max AED 50/day.

- **Model D — Hybrid**

Some credits are shared while others are individually allocated.

Configure:

- Wallet model
- Ownership type
- Maximum members
- Shared balance allowed
- Individual allocations allowed
- Hybrid mode
- Transfer capability
- Credit sharing
- Default permissions
- Default spending policy
- Applicable credit types

Acceptance Condition:

TICVAI supports multiple reusable shared-wallet models without requiring custom development for each customer or venue.

Screen 03 — Family & Household Structure Configuration

Purpose: Configure family relationships and wallet participation.

The source requires customers to register individually or as a **family/group**, with a designated owner able to allocate budgets to linked accounts and monitor/cancel spending authorization.

- **Supported Roles**
- Family Owner
- Co-Owner
- Parent
- Guardian
- Adult Member
- Child
- Dependant
- Viewer

Configure:

- Primary account holder
- Secondary parent/guardian
- Dependents
- Relationship type
- Maximum family members
- Member invitation
- Member verification
- Approval required to join
- Removal rules
- Member status

- Effective dates

Family structure example:

- **Al Qawasmi Family Wallet**
- **Primary Owner**
→ Parent 1
- **Co-Owner**
→ Parent 2
- **Dependants**
→ Child A
→ Child B
→ Child C

Each family member can have independent:

- Credential
- Wristband
- QR
- NFC identifier
- Spending permissions
- Spending limits

Acceptance Condition:

Authorized users can create and maintain family structures while preserving individual identities and wallet permissions.

Screen 04 — Parent–Child Stored Value Distribution

Purpose: Configure the amusement-park style parent-card/child-card wallet model required specifically by **4.3.38**.

- **Example**
- **Parent Wallet**
- Available Stored Value: **AED 1,000**

Linked:

- **Child Card 01**
Daily limit: AED 100
- **Child Card 02**
Daily limit: AED 150
- **Child Wristband 03**
Daily limit: AED 75

Configure:

- Parent wallet
- Child/dependant account
- Child card
- Child wristband
- Maximum linked children
- Shared-value access
- Allocated-value access
- Maximum spend
- Per-transaction limit
- Daily limit
- Venue restriction
- Product/category restriction
- Time restriction
- Credit-type restriction

Parent may:

- Add child
- Remove child
- Increase allowance
- Decrease allowance
- Pause spending
- Resume spending
- Transfer budget
- View transactions

Acceptance Condition:

Multiple child credentials can securely consume configured stored value from a parent-controlled wallet while respecting individual restrictions.

Screen 05 — Allowance & Budget Allocation Engine

Purpose: Configure how wallet owners distribute value to linked members.

- **Allocation Types**
- One-time allowance
- Daily allowance
- Weekly allowance
- Monthly allowance
- Event allowance
- Attraction allowance
- Meal allowance
- Ride allowance
- Shopping allowance
- Custom allocation

Example:

- **Family Wallet Balance: AED 2,000**

Member	Allocation	Limit
Parent	Shared	Unlimited
Child A	AED 250	AED 100/day
Child B	AED 200	AED 75/day
Child C	AED 150	AED 50/day

Configure:

- Allocation amount
- Credit type
- Start date
- End date
- Frequency
- Auto-renewal
- Unused-balance behavior
- Carry-forward
- Return-to-parent
- Expiry
- Reallocation
- Funding priority

Support scheduled allocations such as:

Every Friday → Child receives AED 100.

Acceptance Condition:

Wallet owners or authorized administrators can distribute stored value according to configurable allocation rules while preserving central financial control.

Screen 06 — Member Spending Controls & Permissions

Purpose: Give wallet owners granular control over what each linked member can do.

The source explicitly requires family owners to cap expenses and cancel spending authorization at any time.

- **Permission Controls**

For each member configure:

- View balance
- Spend
- Top up
- Receive funds
- Transfer funds
- Send funds
- Redeem vouchers
- Use gift cards
- Use membership benefits
- Add payment method
- Add/remove wearable
- View transaction history
- **Spending Controls**

Configure:

- Maximum transaction amount
- Hourly limit
- Daily limit
- Weekly limit
- Monthly limit
- Number of transactions
- Product/category restrictions
- Venue restrictions
- Channel restrictions
- Time restrictions

Example:

- **Child A**
- ✓ Rides
- ✓ F&B
- ✓ Attractions
- ✗ Retail
- ✗ Wallet Transfer
- ✗ Top-Up
- Daily limit: **AED 150**
- Single transaction: **AED 50**

Acceptance Condition:

Every member transaction is evaluated against that member's current permissions and spending policy before wallet value is authorized.

Screen 07 — Corporate Wallet & Organizational Hierarchy

Purpose: Configure wallets for companies, schools, partners and other organizations.

Requirement 4.3.27 calls specifically for corporate wallets with spending limits, user-level permissions, departmental allocation and reporting.

- **Organizational Structure**

Example:

- **ABC Corporation**
- Corporate Wallet
AED 100,000
- ↓
- Marketing Department
AED 20,000
- ↓
- Operations
AED 30,000
- ↓
- HR
AED 15,000
- ↓
- Executive
AED 35,000

Then allocate to individual users.

Configure:

- Organization
- Business account
- Corporate wallet
- Cost center
- Department
- Team
- Employee/user
- Corporate hierarchy
- Budget owner
- Finance approver
- Wallet administrator

Support:

- **Corporate → Department → Team → Individual**

Acceptance Condition:

TICVAI can represent hierarchical corporate wallet structures and allocate spending authority at every configured organizational level.

Screen 08 — Corporate Budget, Policy & Approval Rules

Purpose: Control how organizational wallet funds may be spent.

- **Budget Configuration**

Configure:

- Corporate budget
- Department budget
- User budget
- Period
- Currency

- Credit type
- Carry-forward
- Remaining-budget treatment
- **Spending Policies**

Restrict by:

- Venue
- Ticket type
- Event
- Attraction
- F&B
- Retail
- Parking
- Rental
- Product category
- Date/time
- Channel
- **Approval Rules**

Example:

Employee purchase:

- **≤ AED 500**
→ Automatically allowed
- **AED 501–2,000**
→ Manager approval
- **> AED 2,000**
→ Finance approval

Support:

- Pre-approved spending
- Post-spend review
- Single approval
- Multi-level approval
- Cost-center authorization
- Exception approval

Acceptance Condition:

Corporate spending follows configured budgets, policies and approval hierarchies before wallet funds are committed.

Screen 09 — Shared Wallet Transfers & Balance Reallocation

Purpose: Manage movement of stored value within a family or corporate wallet structure.

- **Supported Operations**
- **Family**
 - Parent → Child
 - Child → Parent
 - Parent → Parent
 - Child → Child (*if permitted*)
- **Corporate**

Corporate → Department
 Department → Employee
 Employee → Department
 Department → Corporate
 Department → Department

Configure:

- Transfer permission
- Transferable credit types
- Minimum transfer
- Maximum transfer
- Daily transfer limit
- Approval requirement
- Transfer fees
- Expiry inheritance
- Source-credit preservation
- Return unused allocation
- Automatic reallocation

Example:**Child A has:**

- **AED 80 unused allowance**

Parent chooses:

- **Return AED 80 to Family Wallet**

The ledger records:

- Child allocation – AED 80

Family available balance + AED 80.

No original transaction is deleted.

Acceptance Condition:

All internal balance reallocations are controlled, ledged and traceable to their source and destination.

Screen 10 — Shared Wallet Simulator, Monitoring & Audit

Purpose: Test complex family/corporate wallet policies before publication and provide complete traceability after activation.

- **Simulation Example**
- **Family Wallet**
- Balance: AED 1,500
- **Child A**
- Daily Limit: AED 100
- Used Today: AED 70
- Requested Transaction: AED 50

TICVAI evaluates:

- Wallet Balance → **PASS**
- Member Permission → **PASS**
- Product Eligibility → **PASS**
- Daily Limit → **FAIL**
- Remaining allowance = AED 30
- **Result**
- **Transaction Declined**

Reason: Daily member spending limit exceeded.

- **Corporate Simulation**

Employee requests:

- **AED 1,250 Attraction Package**

TICVAI evaluates:

Corporate balance → PASS

Department budget → PASS

Employee allowance → PASS

Product eligibility → PASS

Approval threshold → **Manager Approval Required**

- **Result**
- **Pending Approval**

-
- **Audit History**

Record:

- Wallet
- Member
- Owner
- Transaction
- Allocation
- Transfer
- Limit change
- Permission change
- Member added/removed
- Credential added/removed
- Suspension
- Reactivation
- Approval
- User performing action
- Old value
- New value
- Date/time
- Reason

Provide:

- Save Draft
- Validate
- Simulate
- Compare Version
- Submit for Approval
- Approve
- Publish
- Roll Back

AI Capability:

AI can detect conflicting shared-wallet rules, excessive permissions, abnormal dependent spending, unusual corporate allocations and policies that may unintentionally prevent legitimate transactions.

Acceptance Condition:

Shared-wallet policies can be simulated before publication, and every configuration or financial action remains traceable through immutable audit history.

Board 4 — Backend Logic

The overall relationship should be:

- Master Wallet / Account
- → Ownership Structure
- → Members / Dependants / Employees
- → Credentials & Wearables
- → Shared or Allocated Balance
- → Individual Permissions
- → Spending Limits
- → Product / Venue Restrictions
- → Approval Rules
- → Transaction
- → Wallet Ledger & Audit

This board is particularly important for TICVAI because it allows the same wallet engine to serve both **theme-park families** and **B2B/corporate clients** rather than developing two separate wallet systems.

It also fully separates the responsibilities of Boards 1–4: **Board 1 defines the wallet, Board 2 funds it, Board 3 determines how credits are consumed, and Board 4 determines how wallet value is shared and controlled among multiple people or organizational users.**

Board 5 — Gift Cards, Vouchers & Digital Benefits

10 Backend Configuration Screens

BOARD 5 – GIFT CARDS, VOUCHERS & DIGITAL BENEFITS

Configure gift cards, digital vouchers, coupons, membership benefits and all digital instruments stored in the TICVAI Wallet.

KPIs:

- Outstanding Liability: AED 18,765,432.50 (↑ 12.8% vs last month)
- Gift Cards Sold: 12,845 (↑ 15.6%)
- Gift Cards Redeemed: 8,932 (↑ 11.3%)
- Active Vouchers: 5,642 (↑ 9.7%)
- Vouchers Redeemed: 3,789 (↑ 13.2%)
- Credits Expiring Soon: 1,245 (↑ 8.8%)
- Pending Activations: 236 (View All)

01 Gift Card & Digital Benefit Command Center

Instrument Overview (AED)

Instrument	Type	Value (AED)	Expiry Date	Count
Summer Promo Voucher	Voucher	245,000.00	15 May 2025	1,345
Gift Card	Gift Card	188,450.00	30 May 2025	876
Gold Member 148 Credit	Credit	152,300.00	30 May 2025	152
Rose Gold Park	Credit	95,700.00	30 May 2025	421

02 Gift Card Product Configuration

Product Name: Theme Park Gift Card

Card Type: Variable Value

Value Range (AED): 50 to 5,000.00

Currency: AED - UAE Dirham

Partial Redemption: ☒ On

Refundable: ☒ On

Expiry Policy: 2 Years from Activation

Applicable Venues: All Theme Parks (3)

Applicable Channels: POS, Web, App, Kiosk

Status: Active

03 Gift Card Issuance, Activation & Distribution

Issuance Source: B2C Website

Activation Method: Activate on Payment

Delivery Method: Email

Sender Name: John Doe

Recipient Email: jdoe@example.com

Message (Optional): Enjoy your gift. Have a great time.

Value (AED): 250.00

Sent Date: 01 May 2025

Allow Recipient to Load to Wallet: ☒ On

04 Voucher & Coupon Type Configuration

Voucher Type	Category	Value Type	Usage	Status
Meal Voucher	F&B	Fixed Amount	Single Use	Active
Parking Voucher	Parking	Fixed Amount	Single Use	Active
Ride Voucher	Attractions	Fixed Amount	Multi Use	Active
Retail Discount	Retail	Percentage	Single Use	Active
Upgrade Voucher	Ticketing	Fixed Amount	Single Use	Active
Complimentary Ticket	Ticketing	Quantity	Single Use	Active
Birthday Reward	Campaign	Fixed Amount	Single Use	Active
Service Recovery	Campaign	Fixed Amount	Single Use	Active

05 Voucher Eligibility & Redemption Rules

Eligibility	Redemption Rules	Restrictions
Voucher Type: Meal Voucher	Applicable Venues: F&B Outlets (40)	Applicable Categories: Food, Beverages
Minimum Spend (AED): 50.00	Valid From: 01 May 2025	Valid Until: 31 Dec 2025
Days of Week: Every Day	Time Window: 11:00 AM - 10:00 PM	Stackable with Promotions: ☒ On
Stackable with Membership Benefits: ☒ On		

06 Membership Benefits & Entitlement Mapping

Benefits	Issuance Triggers	Mapping
Membership Tier	Benefit Type	Value / Qty
Gold	F&B Credit	200.00 AED
Gold	Guest Pass	5
Gold	Parking Credit	2
Gold	Ride Credit	3
Platinum	F&B Credit	400.00 AED
Platinum	VIP Upgrade Voucher	1
Platinum	Birthday Voucher	1
Platinum	Complimentary Ticket	2

07 Benefit Packaging & Wallet Presentation

Category Display Order:

- Money & Balances
- Gift Cards
- Vouchers
- Membership Benefits
- Rewards & Promotions

08 Expiry Management & Notifications

Instrument Type	Expiry Policy	Grace Period	Notify Before	Status
Gift Card	2 Years	90 Days	30, 14, 7, 1 Days	Active
Meal Voucher	90 Days	7 Days	14, 7, 1 Days	Active
Parking Voucher	180 Days	7 Days	14, 7, 1 Days	Active
Member Benefits	End of Membership	0 Days	14, 7, 1 Days	Active
Promo Credits	60 Days	7 Days	7, 1 Days	Active

09 Liability, Breakage & Accounting Control

Liability Summary (AED)

Category	Value
Total Issued Value	24,580,000.00
Outstanding Balance	18,765,432.50
Redeemed Value	4,215,000.00
Expired Value	1,345,482.50
Suspended Value	334,130.00
Breakage Recognized	987,654.30

10 Simulator, Validation & Publication

Simulation Scenario

Item	Amount (AED)	Status
Gift Card (250.00 Balance)	250.00	Used
Meal Voucher (50.00)	50.00	Used
Membership Discount (20%)	35.00	Used
Remaining Amount	65.00	Pending

Board Objective: Configure the complete lifecycle of gift cards, digital vouchers, coupons, campaign rewards, complimentary credits and membership-related benefits stored or presented inside the TICVAI Wallet.

This board covers the requirements to support **gift-card balances, digital vouchers, membership-related credits and benefits, activation, partial redemption, expiry, balance inquiry, transaction history, gift-card liability, and integration with other TICVAI modules.**

The key architecture principle is:

Wallet owns the stored balance and ledger.

Gift Card/Voucher configuration defines the instrument and its rules.

Membership/CRM/Promotions can issue benefits.

Ticketing, F&B, Retail, Parking, Rental and Attractions can redeem them.

Screen 01 — Gift Card & Digital Benefit Command Center

Purpose:

Provide administrators with a consolidated operational view of gift cards, vouchers and digital benefits across all tenants and venues.

- Dashboard KPIs

Display:

- Active Gift Cards
- Outstanding Gift Card Value
- Gift Cards Sold
- Gift Cards Redeemed
- Partial Balances
- Expiring Gift Cards
- Active Vouchers

- Vouchers Redeemed
- Expiring Vouchers
- Membership Benefits Issued
- Membership Benefits Redeemed
- Promotional Credits Outstanding
- **Breakdown By**
 - Gift card type
 - Voucher type
 - Benefit type
 - Tenant
 - Venue
 - Currency
 - Channel
 - Campaign
 - Membership tier
- **Status**
- **Operational Alerts**

Highlight:

- High-value unredeemed gift cards
- Expiring value
- Failed activations
- Redemption exceptions
- Duplicate voucher attempts
- Suspicious redemption activity
- Unusually high manual issuance
- Outstanding liability

Quick Actions

- Create Gift Card
- Create Voucher
- Create Benefit
- Issue Value
- Search Instrument
- Check Balance
- Investigate Redemption
- View Liability

AI Capability:

AI identifies unusual issuance/redemption patterns, abnormal outstanding balances and gift-card or voucher programs with unexpectedly low/high redemption.

Acceptance Condition:

Authorized users can monitor all gift-card, voucher and digital-benefit programs and drill into their balances, issuance and redemption activity.

Screen 02 — Gift Card Product Configuration

Purpose:

Create reusable gift-card products that can be sold or issued across TICVAI.

The source specifically requires gift-card balances including **activation, redemption, partial usage, expiry, balance inquiry and transaction history**.

- **Gift Card Types**

Support:

- Fixed Value
- Variable Value
- Promotional Gift Card
- Corporate Gift Card
- Digital Gift Card
- Physical Gift Card
- Event Gift Card
- Venue Gift Card
- Multi-Venue Gift Card
- **Configuration**

For each gift card define:

- Product name
- Internal code
- Description
- Fixed / variable value
- Minimum value
- Maximum value
- Currency
- Reloadable
- Partial redemption allowed
- Multiple redemption allowed
- Transferable
- Refundable
- Cash-out permitted
- Expiry policy
- Applicable tenant
- Applicable venues
- Applicable channels
- Applicable products
- Activation method
- Wallet storage allowed
- Customer registration required
- Gift recipient details required
- Effective dates
- **Status**
- **Lifecycle**
- **Draft → Approved → Published → Active → Suspended → Retired**

Acceptance Condition:

Administrators can create and maintain gift-card products entirely through backend configuration.

Screen 03 — Gift Card Issuance, Activation & Distribution

Purpose:

Configure how gift cards are created, sold, activated and delivered.

- **Issuance Sources**
- B2C Website
- Mobile App
- POS

- Kiosk
- Customer Service
- Corporate Portal
- B2B/Reseller
- Campaign
- Administrative issuance
- API
- **Activation Methods**
- Activate on sale
- Activate on payment confirmation
- Manual activation
- Scheduled activation
- API activation
- Bulk activation
- **Distribution**

Support:

- Digital delivery
- Email
- SMS
- Mobile app
- QR code
- Physical card
- RFID/NFC card
- Wallet
- Printable voucher

Configure:

- Card number format
- PIN/security code
- Activation trigger
- Recipient
- Sender
- Personal message
- Delivery date
- Delivery channel
- Activation validity
- Resend rules
- Replacement rules
- **Corporate Bulk Issuance**

Allow organizations to create:

- 500 × AED 100 gift cards

with controlled:

- Batch number
- Activation
- Distribution
- Expiry
- Cost center
- Reporting

Acceptance Condition:

Every issued gift card has a unique identity, activation state, value, distribution record and complete audit history.

Screen 04 — Voucher & Coupon Type Configuration

Purpose:

Create and manage digital vouchers and non-cash redemption instruments.

The source requires the wallet to store and manage **digital vouchers, coupons, meal vouchers, parking vouchers, attraction credits and campaign rewards**.

- **Voucher Types**
- Monetary Voucher
- Percentage Voucher
- Complimentary Ticket
- Meal Voucher
- F&B Voucher
- Retail Voucher
- Parking Voucher
- Attraction Voucher
- Ride Voucher
- Rental Voucher
- Upgrade Voucher
- Campaign Reward
- Service Recovery Voucher
- Custom Voucher

Configure

- Voucher name
- Voucher code
- Voucher category
- Monetary/non-monetary
- Fixed value
- Percentage value
- Quantity entitlement
- Applicable products
- Applicable categories
- Applicable venues
- Applicable channels
- Minimum spend
- Maximum benefit
- Validity
- Single-use/multi-use
- Partial redemption
- Stackable
- Transferable
- Customer-specific
- Wallet storage
- Barcode / QR

- **Status**

Acceptance Condition:

New voucher types and redemption behaviors can be introduced without development changes.

Screen 05 — Voucher Eligibility & Redemption Rule Studio

Purpose:

Define precisely when, where and by whom a voucher may be redeemed.

- **Eligibility Rules**

Configure by:

- **Customer**
 - Customer segment
 - Membership tier
 - Age
 - Family role
 - Corporate account
- **Product**
 - Ticket
 - Attraction
 - Ride
 - F&B
 - Retail
 - Rental
 - Parking
 - Add-on
- **Location**
 - Tenant
 - Venue
 - Outlet
 - Attraction
- **Channel**
 - POS
 - B2C
 - Mobile App
 - Kiosk
 - API
- **Time**
 - Valid dates
 - Days of week
 - Time window
 - Season
 - Event period
- **Redemption Controls**

Configure:

- Maximum redemptions
- Per-customer limit
- Per-transaction limit
- Minimum basket value
- Partial redemption
- Combination with promotions
- Combination with membership discounts
- Combination with wallet credits
- Excluded products

- **Example**
- **Family Meal Voucher**

Valid:

- ✓ F&B
- ✓ Theme Park Venue
- ✓ 11:00–18:00
- ✓ One per family/day

Not valid:

- ✗ Retail
- ✗ Parking
- ✗ Cash conversion

Acceptance Condition:

Every voucher redemption is validated in real time against the currently published eligibility and redemption policy.

Screen 06 — Membership Benefits & Entitlement Mapping

Purpose:

Connect Membership & Loyalty benefits into the Wallet without making Wallet responsible for membership logic.

Requirement 4.3.25 requires membership-related **benefits, credits, entitlements, guest passes, reservation quotas and included benefits** to be available within the guest wallet.

- **Membership Benefit Examples**
- **Gold Membership**

→ AED 200 F&B Credit

→ 5 Guest Passes

→ 2 Parking Credits

→ 3 Complimentary Ride Credits

→ 1 Birthday Voucher

- **Configure Mapping**
- **Membership Product**
 - Membership Tier
- → Benefit Type
- → Wallet Credit/Voucher/Entitlement

Configure:

- Membership tier
- Benefit
- Quantity/value
- Issuance timing
- Renewal behavior
- Validity
- Usage frequency
- Applicable venue
- Applicable product
- Carry-forward
- Transferability
- Family sharing
- Replacement rules
- **Issuance Triggers**
- Membership activation
- Membership renewal
- Tier upgrade

- Anniversary
- Birthday
- Visit milestone
- Campaign
- Manual award

Acceptance Condition:

Membership-generated benefits are automatically represented in the customer's wallet according to their configured entitlement rules while Membership remains the authoritative source.

Screen 07 — Benefit Packaging & Digital Wallet Presentation

Purpose:

Configure how different instruments appear to customers in B2C and mobile wallet experiences.

- **Wallet Sections**

Administrators can define presentation categories:

- **Money**
 - Cash Balance
 - Refund Credit
- **Gift Cards**
 - Gift Card Balance
- **Vouchers**
 - Meal Voucher
 - Parking Voucher
 - Attraction Voucher
- **Membership**
 - Guest Passes
 - Complimentary Tickets
 - Member Credits
- **Rewards**
 - Promotional Credits
 - Campaign Rewards

Configure

- Display name
- Customer description
- Icon
- Category
- Balance visibility
- Expiry visibility
- Terms link
- QR visibility
- Barcode visibility
- Redeem button
- Transfer button
- Gift button
- Applicable channels
- Display priority

Configure states:

- Available
- Reserved

- Partially Used
- Used
- Expiring Soon
- Expired
- Suspended

Acceptance Condition:

Backend administrators can determine how wallet instruments and benefits are categorized and presented consistently across supported customer channels.

Screen 08 — Gift Card & Voucher Expiry Management

Purpose:

Control validity, expiration, extension and customer notification.

- **Expiry Models**
- Never expires
- Fixed date
- X days after purchase
- X days after activation
- X days after issuance
- End of campaign
- End of event
- Membership expiry

Configure:

- Valid-from
- Valid-until
- Grace period
- Extension allowed
- Maximum extension
- Manual extension
- Renewal
- Remaining-balance behavior
- Expired-value treatment
- **Notification Schedule**

The system can notify:

- **30 days before**
- **14 days before**
- **7 days before**
- **1 day before**
- **On expiry**

Channels:

- Push
- Email
- SMS
- In-app
- Configured messaging integrations
- **Administrative Actions**

Authorized users can:

- Extend
- Reinstate

- Suspend
- Cancel
- Replace
- Reissue

Every action requires appropriate permissions and audit history.

Acceptance Condition:

Gift cards and vouchers follow configurable validity policies and customers receive applicable expiry notifications.

Screen 09 — Gift Card Balance, Liability & Breakage Control

Purpose:

Connect gift-card operations to financial liability management without duplicating the Finance module.

The source explicitly requires **Gift Card Liability Reporting** and **Gift Card Breakage & Revenue Recognition**.

- **Liability Dashboard**

Display:

- Gift cards sold
- Original issued value
- Outstanding balance
- Redeemed balance
- Expired balance
- Suspended balance
- Unredeemed balance
- Breakage amount
- Recognized revenue
- Deferred liability

Break down by:

- Tenant
- Venue
- Currency
- Gift-card program
- Issuance period
- Accounting period
- **Breakage Policy**

Configure:

- Expiration treatment
- Breakage calculation method
- Recognition timing
- Accounting mapping
- Deferred revenue account
- Revenue account
- Liability account
- Tax treatment
- Finance approval
- **Important Architecture**
- **Wallet Module**

maintains:

Individual balance + ledger.

- **Finance Module**

owns:

Accounting entries + financial reporting.

This screen provides the wallet-side configuration and operational visibility required to connect both domains.

Acceptance Condition:

Outstanding gift-card balances can be reconciled to financial liability, and configured expiry/breakage events produce traceable finance integration events.

Screen 10 — Gift Card & Voucher Simulator, Validation & Publication

Purpose:

Allow administrators to test complete gift-card, voucher and benefit configurations before activation.

- **Scenario Example**
- **Customer:** Gold Member
- **Venue:** Theme Park

Basket:

- 2 Adult Tickets — AED 400
Lunch — AED 120
Parking — AED 50

Available Wallet Instruments:

- Gift Card → AED 200
- Meal Voucher → AED 50
- Parking Voucher → 1 Free Parking
- Membership Benefit → 10% Ticket Discount
- **TICVAI Simulation**
- **Membership Discount**
AED 40
- **Gift Card**
AED 200
- **Meal Voucher**
AED 50
- **Parking Voucher**
AED 50
- **Remaining Amount**
AED 230

→ External payment required.

- **Rule Evaluation**
- Membership Eligibility → **PASS**
- Gift Card Validity → **PASS**
- Gift Card Product Eligibility → **PASS**
- Meal Voucher Eligibility → **PASS**
- Parking Voucher → **PASS**
- Expiry → **PASS**
- Stacking Rules → **PASS**
- **Administrator Can Inspect**
- Rule executed
- Instrument selected
- Rejected instruments
- Rejection reason
- Redemption amount
- Remaining balance
- Liability impact

- Wallet ledger impact
- **Publication Controls**
- Save Draft
- Validate
- Simulate
- **Compare Versions**
- Submit for Approval
- Approve
- Schedule Publication
- Publish
- Suspend
- Roll Back

AI Capability:

AI identifies conflicting eligibility rules, potentially unusable vouchers, incorrect benefit mappings, excessive liability exposure and configuration combinations likely to cause redemption failures.

Acceptance Condition:

No new gift-card, voucher or benefit configuration becomes operational until it passes system validation and applicable approval/publication controls.

Board 5 — Backend Architecture Flow

The backend should operate approximately as:

- Gift Card / Voucher / Benefit Definition
- → Issuance Rule
- → Activation
- → Wallet Association
- → Eligibility
- → Validity & Expiry
- → Redemption
- → Partial Balance / Remaining Entitlement
- → Wallet Ledger
- → Liability / Finance Event
- → Audit

This keeps the modules clean:

Board 1: Wallet foundation

Board 2: Funding & top-up

Board 3: Stored value & consumption

Board 4: Family & corporate sharing

Board 5: Gift cards, vouchers & digital benefits

The next board should be **Board 6 — Wallet Payments, Wearables & Channel Usage**. That board will cover how the wallet is actually used across **POS, B2C, mobile app, kiosks, RFID wristbands, NFC, QR, attractions, F&B and Retail**, including credential-to-wallet resolution and online/offline transaction behavior. This maps particularly strongly to requirements

4.3.7, 4.3.8, 4.3.10, 4.3.12 and 4.3.13

- RFID Transactions
- Declined Wallet Transactions
- Offline Transactions
- Pending Synchronizations
- **Usage by Business Area**

Break down by:

- Ticketing
- Attractions
- Rides
- F&B
- Retail
- Rental
- Parking
- Events
- Membership
- Other services

Filters

- Tenant
- Venue
- Channel
- Outlet
- Device
- Wallet type
- Credential type
- Transaction status
- Currency
- Date/time
- **Operational Alerts**

Highlight:

- High decline rate
- Offline devices
- Credential failures
- Duplicate payment attempts
- Unsynchronized transactions
- Abnormal wearable activity
- Wallet-service latency

AI Capability:

AI identifies unusual wallet usage patterns and predicts channel/device issues before they materially impact operations.

Acceptance Condition:

Authorized users can monitor wallet consumption across channels and drill down to individual wallet transactions and devices.

Screen 02 — Wallet Channel Configuration

Purpose: Define which TICVAI channels can use Wallet as a payment or redemption method.

Requirement 4.3.8 requires multiple wallet payment channels including mobile applications and wearables, with online, onsite POS and in-app payment modes.

- **Supported Channels**

Configure:

- B2C Website
- Customer Mobile App
- POS
- Mobile POS
- Kiosk
- Self-Service Terminal
- F&B POS
- Retail POS
- Parking
- Attraction Terminal
- Access Device
- Customer Service
- Corporate Portal
- B2B
- API
- Third-party system
- **Per Channel Configure**
- Wallet payment enabled
- Balance inquiry enabled
- Voucher redemption
- Gift card redemption
- Refund-to-wallet
- Split tender
- Top-up
- Transfer
- Offline wallet usage
- Supported currencies
- Supported credit types
- Maximum transaction
- Authentication requirement
- Credential types
- Applicable tenants/venues

Support configuration inheritance:

- Global → Tenant → Venue → Channel → Device

Acceptance Condition:

Wallet capabilities are available only through explicitly configured and authorized channels.

Screen 03 — Wallet Payment & Redemption Policy

Purpose: Configure the business rules applied when a customer selects Wallet at checkout.

Configure

- Wallet payment enabled
- Full wallet payment
- Partial wallet payment
- Split tender
- Minimum wallet contribution
- Maximum wallet contribution
- Maximum transaction value
- Minimum remaining balance

- Negative balance allowed/not allowed
- External tender fallback
- PIN requirement
- MFA requirement
- Customer confirmation
- Receipt requirement
- **Transaction Types**

Policies can differ for:

- Ticket purchase
- F&B
- Retail
- Attraction
- Ride
- Parking
- Rental
- Membership
- Add-on
- **Example**
- **F&B Purchase: AED 150**
- Available wallet: AED 100

Policy:

- Partial wallet payment = Allowed

Result:

- Wallet → AED 100
Card → AED 50

Whereas another venue could configure:

- Partial wallet payment = Not Allowed

Result:

Wallet payment rejected and another tender requested.

Acceptance Condition:

Every wallet payment follows the applicable published transaction and redemption policy.

Screen 04 — Wearable & Credential Type Configuration

Purpose: Define physical and digital credentials that can represent a wallet.

The source requires venue-provided wearables to be linked to digital wallets and used to make payments.

- **Credential Types**

Support:

- RFID Wristband
- NFC Wristband
- RFID Card
- NFC Card
- QR Code
- Dynamic QR
- Mobile Wallet Credential
- Digital Key
- Employee Credential
- Hotel Key

- Event Badge
- Custom Credential

Configure

- Credential name
- Internal code
- Technology
- Identifier format
- Tokenization
- Encryption requirement
- Wallet linking allowed
- Multiple credentials per wallet
- Credential sharing permitted
- PIN required
- Customer verification
- Activation
- Expiry
- Replacement
- Lost/stolen handling
- Offline eligibility
- Applicable venue
- Applicable device types

Acceptance Condition:

Administrators can introduce and govern wallet credentials independently from the wallet account itself.

Screen 05 — Wearable Linking & Wallet Association Rules

Purpose: Configure how wristbands, cards and other credentials are linked to wallet accounts.

Requirement 4.3.12 specifically requires guests to link and remove venue-issued wearables from their wallet.

- **Linking Methods**
- Customer Mobile App
- B2C Portal
- POS
- Guest Services
- Kiosk
- Hotel Check-In
- Event Registration
- Administrative Backend
- API

Configure

- Wallet type eligibility
- Credential type
- Customer verification
- Activation code
- PIN verification
- OTP verification
- Digital ID verification
- Maximum credentials per wallet
- Maximum wallets per credential
- Automatic activation

- Manual approval
- Effective date
- Expiry
- **Credential Lifecycle**
- **Unassigned → Linked → Active → Suspended → Lost/Stolen → Replaced → Unlinked → Retired**
- **Administrative Actions**
- Link
- Unlink
- Suspend
- Reactivate
- Replace
- Transfer
- Mark Lost
- Mark Stolen
- View Usage

When replacing a wristband:

- **Old credential is disabled**
- →
- **New credential links to same wallet**
- →
- **Wallet balance remains unchanged**

Acceptance Condition:

Credentials can be securely linked, removed or replaced without modifying the underlying wallet balance or ledger.

Screen 06 — NFC, RFID & QR Interaction Rules

Purpose: Configure how different proximity/contactless technologies interact with TICVAI Wallet.

Requirement 4.3.13 specifically requires wallet payments using **RFID, NFC and QR code**.

- **Configure by Technology**
- **RFID**
- Reader type
- Card/wristband format
- Identifier validation
- Offline support
- **NFC**
- Token validation
- Device authentication
- Tap behavior
- Secure token requirements
- **QR**
- Static QR
- Dynamic QR
- QR validity period
- Single-use token
- Refresh frequency
- Screenshot protection controls where supported
- **Interaction Types**

Configure:

- Tap to Pay

- Scan to Pay
- Tap to Redeem
- Scan to Redeem
- Balance Check
- Attraction Entry
- Ride Redemption
- F&B Purchase
- Retail Purchase
- Parking
- **Transaction Feedback**

Device response:

- **Green**
→ Approved
- **Red**
→ Declined
- **Amber**
→ Additional action required

Backend response includes:

- Authorization result
- Amount
- Remaining balance
- Decline reason
- Credential status

Acceptance Condition:

All supported proximity interactions resolve securely to the correct wallet and enforce the applicable wallet policy.

Screen 07 — Digital Key & Wallet Authentication Policy

Purpose: Configure authentication requirements based on transaction risk.

Requirement 4.3.10 specifically calls for **Digital Key/Digital ID authentication wherever possible** to enhance security and credential management.

- **Authentication Methods**

Support:

- Wallet PIN
- Device authentication
- OTP
- Mobile biometric authentication
- Digital Key
- Digital ID
- Customer login
- Membership credential
- Staff authorization
- Step-up authentication
- **Risk-Based Rules**

Example:

- **≤ AED 50**
→ Tap and Pay
- **AED 51–250**
→ Device confirmation

- **AED 251–1,000**
→ PIN / biometric
- **> AED 1,000**
→ Strong authentication

Configure based on:

- Transaction amount
- Credit type
- Customer
- Wallet type
- Venue
- Channel
- Credential
- Risk score
- Transaction velocity

Acceptance Condition:

TICVAI dynamically applies the required authentication method before wallet authorization according to published risk rules.

Screen 08 — Offline Wallet & Degraded Mode Configuration

Purpose: Maintain controlled venue operations during temporary network or service interruptions.

This is especially important for amusement parks, festivals, stadiums and large attractions where wristband transactions cannot simply stop because connectivity is temporarily unavailable.

- **Offline Eligibility**

Configure by:

- Venue
- Outlet
- Device
- Credential
- Wallet type
- Credit type
- Transaction type
- **Offline Controls**

Configure:

- Offline wallet enabled
- Maximum offline transaction
- Maximum cumulative offline spend
- Maximum transactions
- Offline duration
- Cached balance permitted
- Reserved offline allowance
- Credential whitelist
- Risk profile
- Expiry of cached data
- Device storage requirements
- **Example**

Child Wristband:

- Wallet balance before disconnection:
AED 300

- Offline allowance:
AED 100

The child may continue spending up to:

- **AED 100 offline**

regardless of the previously cached AED 300 balance.

This limits financial exposure.

- **Recovery**

When connectivity returns:

- **Device Transactions**
- → Upload
- → Validate
- → Reconcile
- → Wallet Ledger Update
- → Exception Handling

Acceptance Condition:

Offline wallet usage remains within configured exposure limits, and all transactions synchronize and reconcile after connectivity restoration.

Screen 09 — Device, Terminal & Acceptance Point Mapping

Purpose: Determine exactly which devices and locations are authorized to accept TICVAI Wallet.

- **Hierarchy**
- **Tenant**
- → Venue
- → Business Area
- → Outlet / Attraction
- → Terminal
- → Device
- **Acceptance Points**

Examples:

- Main Ticket POS
- Restaurant POS
- Retail POS
- Ride Reader
- Attraction Reader
- Parking Terminal
- Kiosk
- Mobile POS
- Handheld Device
- Turnstile
- Self-Service Terminal

Configure

- Device ID
- Device type
- Venue
- Outlet
- Business function
- Wallet enabled
- Supported credentials

- Supported credit types
- Maximum transaction
- Offline permission
- Device risk profile
- Authentication policy
- Operating hours
- Device status

Statutes:

- **Active / Offline / Maintenance / Suspended / Retired**
- **Device-Level Overrides**

Example:

- **Ride Reader #R102**

Allowed:

- ✓ Ride Credits
- ✓ Membership Ride Benefits

Not allowed:

- ✗ Cash Credit
- ✗ Gift Card
- ✗ F&B Credit

Acceptance Condition:

Only registered and authorized acceptance points can initiate wallet payment or redemption requests.

Screen 10 — Wallet Transaction Simulator, Monitoring & Channel Audit

Purpose: Test the complete channel-to-wallet transaction flow before configuration is published.

- **Simulation Scenario**
- **Customer:** Family Member / Child
- **Wallet:** Family Wallet
- **Credential:** RFID Wristband #CH-0045
- **Venue:** Theme Park
- **Device:** F&B POS #12
- **Purchase:** AED 85
- **TICVAI Evaluation**
- Credential Active → **PASS**
- Credential Linked → **PASS**
- Device Authorized → **PASS**
- Channel Wallet Enabled → **PASS**
- F&B Credit Eligible → **PASS**
- Member Spending Permission → **PASS**
- Daily Limit → **PASS**
- Authentication Requirement → **PASS**
- Connectivity → **ONLINE**
- **Consumption**
- Meal Credit → AED 30
- Bonus Credit → AED 20
- Cash Credit → AED 35
- **Transaction Approved: AED 85**

Remaining wallet balances displayed.

- **Alternative Failure Scenario**

RFID wristband:

- **Status = LOST**

Result:

- **DECLINED**

Reason:

Credential suspended — lost/stolen status.

- **Monitoring & Audit**

Record:

- Wallet ID
- Credential
- Device
- Channel
- Venue
- Customer/member
- Transaction
- Credit buckets consumed
- Authentication
- Online/offline
- Authorization result
- Decline reason
- Latency
- Date/time
- **Publication Controls**
- Save Draft
- Validate
- Simulate
- Compare Version
- Submit for Approval
- Approve
- Publish
- Roll Back

AI Capability:

AI identifies conflicting channel configurations, unsupported credential combinations, excessive offline exposure and device policies likely to cause customer transaction failures.

Acceptance Condition:

Administrators can simulate wallet transactions end-to-end before publication, and every real transaction remains traceable from the acceptance device through credential resolution, authorization, consumption and wallet ledger posting.

Board 6 — Backend Transaction Flow

This is the important architecture to follow:

- **Customer / Family Member**
- → **Credential**
RFID / NFC / QR / Mobile / Digital Key
- → **Acceptance Point**
POS / Kiosk / Ride / F&B / Retail / Parking / App / B2C
- → **Credential Resolution**
- → **Wallet Identification**
- → **Channel Policy**
- → **Member Permission**
- → **Authentication / Risk Check**
- → **Credit Eligibility**
- → **Consumption Engine from Board 3**
- → **Wallet Authorization**
- → **Debit / Redemption**
- → **Ledger**
- → **Receipt / Device Response**

This gives us a clean progression:

Board 1 — Wallet Foundation**Board 2 — Funding & Top-Up****Board 3 — Stored Value & Consumption****Board 4 — Family & Corporate Wallets****Board 5 — Gift Cards, Vouchers & Benefits****Board 6 — Payments, Wearables & Channel Usage**

For **Board 7**, we should move into **Transfers, Refunds & Wallet Operations**—covering P2P wallet transfers, instant refund-to-wallet, reversals, administrative adjustments, balance corrections, blocking/freezing, transaction disputes and operational exception management. That directly addresses requirements such as **4.3.15, 4.3.16, 4.3.21 and 4.3.35**.

Board 7 — Transfers, Refunds & Wallet Operations

10 Backend Configuration Screens

BOARD 7 - TRANSFERS, REFUNDS & WALLET OPERATIONS

Manage wallet-to-wallet transfers, instant refunds, reversals, adjustments, blocks, disputes and operational exceptions with full audit and approval.

Transfers (Today) 356 ↑ 14.8%
AED 128,450.20

Refunds (Today) 278 ↑ 13.2%
AED 95,230.50

Reversals (Today) 82 ↓ 5.4%
AED 21,845.00

Manual Adjustments 46 ↑ 6.7%
AED 12,350.75

Blocked Wallets 74 ↑ 2.7%
Total Balance AED 86,230.10

Disputes Open 32 ↓ 8.3%
High Priority 6

Operations Pending 18
Awaiting Approval

01 Wallet Operations Command Center

Operations Overview (This Month)

Recent Operations

02 Peer-to-Peer Transfer Configuration

General Limits Fees Security Channels

03 Transfer Eligibility & Approval Rules

Eligibility Approval Matrix Restrictions

04 Refund-to-Wallet Policy Configuration

General Routing Validity Notifications

05 Refund Routing & Credit Restoration Engine

Restoration Strategy Restore Original Credit Buckets

06 Reversal & Transaction Corrections

Reversal Types Rules Reason Codes

07 Administrative Balance Adjustment Studio

Credit Adjustment Debit Adjustment Limits

08 Wallet Block, Freeze & Restriction Management

Block / Freeze Restrictions History

09 Disputes & Operational Exception Queue

Open Cases My Cases Closed Cases

10 Operations Simulator, Approval & Audit Trail

Simulation Audit Trail Approvals

Board Objective: Configure the complete operational lifecycle of wallet value after it has been funded or issued—including **peer-to-peer transfers, refund-to-wallet, reversals, balance adjustments, blocking/freezing, disputes, corrections, exception handling and governed administrative actions.**

This board directly addresses requirements **4.3.15, 4.3.16, 4.3.21 and 4.3.35**, while complementing the family/corporate internal allocation capabilities already configured in Board 4.

A critical distinction is:

- **Board 4:** redistribution inside a shared Family/Corporate structure.
- **Board 7:** operational movement/correction of wallet value, including independent wallet-to-wallet transfers, refunds, reversals and administrative intervention.

Screen 01 — Wallet Operations Command Center

Purpose: Provide operations, finance and customer-service administrators with a centralized view of wallet operational activity and exceptions.

- **Dashboard KPIs**

Display:

- Transfers Today
- Transfer Value
- Refunds Today
- Refund Value
- Reversals
- Manual Adjustments
- Blocked Wallets
- Frozen Balances

- Pending Operations
- Failed Operations
- Disputed Transactions
- Operations Awaiting Approval
- **Activity Breakdown**

By:

- P2P Transfer
- Refund
- Reversal
- Credit Adjustment
- Debit Adjustment
- Wallet Block
- Wallet Unblock
- Balance Freeze
- Correction
- Administrative Operation

Filters

- Tenant
- Venue
- Wallet type
- Customer
- Operation
- Credit type
- Currency
- **Status**
- Risk level
- Administrator
- Date/time
- **Operational Alerts**

Highlight:

- High-value manual adjustments
- Repeated refunds
- Failed reversals
- Unusual P2P transfers
- Blocked wallets with remaining balances
- Operations pending approval
- Aged disputes

AI Capability:

AI prioritizes operational exceptions based on financial value, customer impact, transaction age and risk.

Acceptance Condition:

Authorized users can monitor all wallet operational activity and trace every operation to the underlying wallet, transaction and administrator action.

Screen 02 — Peer-to-Peer Transfer Configuration

Purpose: Configure direct value transfers between independent TICVAI wallet accounts.

Requirement **4.3.16** specifically requires guests to transfer money from their wallet to another guest's wallet.

- **Transfer Methods**

Support:

- Wallet ID
- Mobile number
- Email
- QR code
- Customer ID
- Membership ID
- Contact selection
- NFC interaction
- API

Configure

- P2P enabled
- Eligible wallet types
- Eligible customer types
- Eligible credit types
- Supported currencies
- Minimum transfer
- Maximum transfer
- Daily transfer amount
- Monthly transfer amount
- Maximum transfer count
- Transfer fee
- Recipient verification
- Sender authentication
- Recipient confirmation
- Transfer expiry
- Cross-tenant transfer
- Cross-venue transfer
- Cross-currency transfer
- **Example**
- **Wallet A**
- Cash Credit: AED 500

Transfer:

- **AED 100 → Wallet B**

Result:

- Wallet A → AED 400
- Wallet B → +AED 100

Both ledger entries share the same transfer reference.

Acceptance Condition:

P2P transfers create balanced debit and credit ledger entries and occur only after all eligibility, security and limit rules have passed.

Screen 03 — Transfer Eligibility, Limits & Approval Rules

Purpose: Apply governance controls to wallet-to-wallet transfers.

- **Eligibility**

Configure by:

- Wallet type
- Credit type

- Customer verification status
- Membership
- Customer age
- Family role
- Corporate role
- Tenant
- Venue
- Country/region where applicable
- **Credit Transferability**

Example:

Credit	Transferable
Cash Credit	✓
Refund Credit	Configurable
Bonus Credit	×
Membership Credit	×
Ride Credit	×
Gift Balance	Configurable

- **Limits**

Configure:

- Per transaction
- Per hour
- Daily
- Weekly
- Monthly
- Recipient limits
- Sender limits
- **Approval / Authentication**

Example:

- **AED 1–250**
→ Standard authentication
- **AED 251–1,000**
→ MFA
- **AED 1,001+**
→ MFA + Risk Validation

High-risk administrative transfers can require approval.

Actions

When rule fails:

- Decline
- Hold
- Require MFA
- Require approval
- Risk review
- Temporarily restrict transfers

Acceptance Condition:

Every transfer is evaluated against current eligibility, transferability, velocity and authorization rules before wallet value moves.

Screen 04 — Refund-to-Wallet Policy Configuration

Purpose: Configure when refunds from TICVAI transactions may be returned directly to a digital wallet.

Requirement **4.3.15** specifically requires **instant refund to the digital wallet** with customer notification.

- **Refund Sources**
- Ticket cancellation
- Event cancellation
- F&B refund
- Retail return
- Rental refund
- Parking refund
- Membership adjustment
- Attraction cancellation
- Failed service
- Customer-service compensation
- **Refund Destinations**

Configure:

- Original wallet
- Refund Credit bucket
- Cash Credit
- Original gift-card balance
- Original credit bucket
- External payment method
- **Rules**

Configure:

- Instant wallet refund
- Full refund
- Partial refund
- Refund amount limit
- Refund fee
- Refund validity
- Refund-credit expiry
- Transferability
- Withdrawal eligibility
- Original tender dependency
- Original customer requirement
- Approval requirement
- **Example**

Original transaction:

- AED 300

Paid:

- Wallet → AED 200
Card → AED 100

Refund policy can define:

Wallet → AED 200 returned
Card → AED 100 refunded externally.

Or, where permitted:

Full AED 300 → Wallet Refund Credit.

Acceptance Condition:

Refunds follow configured tender and wallet policies while preserving a direct relationship with the original transaction.

Screen 05 — Refund Routing & Credit Restoration Engine

Purpose: Determine exactly **where returned value goes** when the original transaction consumed multiple wallet credits.

This is important because a simple “+AED 200 to Cash Credit” can corrupt wallet economics.

- **Example**

Original purchase:

- **AED 150**

Consumed:

- Meal Credit → AED 30
- Bonus Credit → AED 40
- Gift Card → AED 50
- Cash Credit → AED 30

Full refund should be capable of restoring:

- Meal Credit → +AED 30
- Bonus Credit → +AED 40
- Gift Card → +AED 50
- Cash Credit → +AED 30

rather than creating AED 150 Cash Credit.

- **Configure Restoration Strategy**
- Restore original credit buckets
- Restore original credit lots
- Create Refund Credit
- Return to gift card
- Return to parent wallet
- Return to corporate budget
- Custom routing
- **Expired Credit Handling**

If original Bonus Credit expired after purchase:

Configure:

- Restore with original expiry
- Restore with new validity
- Convert to refund credit
- Do not restore
- Require administrator approval
- **FEFO Interaction**

Restore:

- Original lot
- Original expiry
- Original issuance reference

where business policy requires it.

Acceptance Condition:

Refund processing correctly reconstructs or converts the value originally consumed without unintentionally changing its financial or promotional classification.

Screen 06 — Reversal & Transaction Correction Management

Purpose: Correct operational wallet transactions while maintaining an immutable ledger.

- **Supported Operations**
- Payment reversal
- Transfer reversal
- Refund reversal
- Duplicate debit correction
- Duplicate credit correction
- Failed transaction correction
- Offline synchronization correction
- Incorrect wallet correction
- Incorrect amount correction
- **Important Rule**

Never delete or overwrite the original wallet ledger transaction.

For example:

Original debit:

- – AED 100

Correction:

- Reversal + AED 100

Then, if required:

Correct debit:

- – AED 80

All three entries remain visible.

Configure

- Eligible transaction type
- Reversal period
- Full reversal
- Partial reversal
- Reason code
- Notes
- Approval
- Supporting documentation
- Customer notification
- Finance notification
- Reconciliation behavior

Acceptance Condition:

Every reversal or correction creates linked compensating ledger entries while preserving the complete original transaction history.

Screen 07 — Administrative Balance Adjustment Studio

Purpose: Allow authorized personnel to perform governed wallet balance adjustments.

- **Adjustment Types**
- **Credit**
- Customer compensation
- Service recovery
- Promotional correction
- Migration adjustment

- Balance correction
- Manual award
- **Debit**
- Incorrect credit recovery
- Duplicate credit recovery
- Administrative correction
- Fraud recovery where authorized

Configure

- Adjustment reason
- Credit type
- Amount
- Wallet
- Currency
- Expiry
- Financial classification
- Customer-visible description
- Supporting reference

Approval Matrix

Example:

- **≤ AED 100**
Supervisor
- **AED 101–1,000**
Operations Manager
- **AED 1,001–10,000**
Finance Manager
- **> AED 10,000**
Finance Director

Support:

- Maker-checker
- Mandatory reason
- Mandatory attachment
- Multi-level approval
- Maximum user authority

AI Capability

AI warns when an adjustment is unusual compared with the administrator's historical actions or the customer's normal wallet activity.

Acceptance Condition:

No manual balance adjustment can occur without the required permissions, reason, approval and ledger record.

Screen 08 — Wallet Block, Freeze & Restriction Management

Purpose: Protect wallet accounts and individual balances without necessarily closing the wallet.

Requirement **4.3.21** explicitly states that it must be possible to block a wallet account.

Administrative Controls

Support:

- Block wallet
- Unblock wallet
- Suspend wallet
- Freeze entire balance

- Freeze specific credit
- Disable spending
- Disable transfers
- Disable top-up
- Disable refund
- Disable wearable
- Disable online use
- Disable API use
- **Reason Codes**
- Customer request
- Lost credential
- Suspected fraud
- Chargeback
- Compliance review
- Security concern
- Duplicate account
- Administrative investigation
- Corporate instruction
- **Duration**

Configure:

- Immediate
- Until date
- Until investigation completed
- Permanent
- Manual release
- **Example**

Wallet:

- **AED 850 total**
- Cash Credit → AED 500
- Bonus → AED 150
- Gift Card → AED 200

Administrator may freeze only:

- **Gift Card AED 200**

while allowing the other AED 650 to remain usable.

Acceptance Condition:

Authorized administrators can apply granular restrictions without modifying or deleting underlying wallet balances.

Screen 09 — Wallet Disputes & Operational Exception Queue

Purpose: Provide a controlled workflow for wallet transaction problems requiring investigation.

- **Exception Types**
- Customer disputes transaction
- Duplicate wallet debit
- Wallet charged but service failed
- Transfer recipient not received
- Refund missing
- Offline transaction conflict
- Balance mismatch
- Gift card issue

- Credential misuse
- Suspicious transfer
- Reconciliation mismatch
- **Case Information**

Capture:

- Case ID
- Wallet
- Customer
- Transaction
- Amount
- Channel
- Device
- Venue
- Date/time
- Issue category
- Customer description
- Evidence
- Risk score

Workflow

- **New**
- → **Assigned**
- → **Under Investigation**
- → **Pending Customer**
- → **Pending Finance**
- → **Resolved**
- → **Closed**
- **Resolution Actions**

Authorized users may:

- No action
- Refund
- Reverse
- Adjust balance
- Block wallet
- Replace credential
- Escalate
- Refer to finance/security
- **SLA**

Configure:

- Priority
- Resolution SLA
- Assignment
- Escalation
- Notifications

AI Capability:

AI groups related exceptions, detects repeated patterns and suggests likely root causes using transaction, device and wallet history.

Acceptance Condition:

Every operational wallet dispute is managed through a traceable case workflow linked to the affected transactions and corrective actions.

Screen 10 — Operations Simulator, Approval & Audit Trail

Purpose: Provide administrators with a final controlled workspace for testing sensitive wallet operations and reviewing the immutable operational history.

- **Simulation Example — Refund**

Original purchase:

- **AED 250**

Funding:

- Bonus Credit → AED 50
Gift Card → AED 100
Cash Credit → AED 100

Requested refund:

- **AED 150**

TICVAI evaluates:

- Original Transaction → **FOUND**
- Refund Eligibility → **PASS**
- Refund Period → **PASS**
- Refund Limit → **PASS**
- Credit Restoration → **PASS**
- Approval → **NOT REQUIRED**
- **Proposed Result**
- Bonus Credit → +AED 50
Gift Card → +AED 100
- Refund Total → AED 150

Administrator sees the proposed ledger entries **before execution**.

- **Simulation Example — P2P Transfer**

Sender balance:

- **AED 600**

Transfer:

- **AED 300**

System evaluates:

Recipient → PASS
Credit Transferability → PASS
Daily Limit → PASS
Velocity → PASS
Authentication → **MFA REQUIRED**

Result:

- **Awaiting Authentication**

- **Audit Trail**

Requirement **4.3.35** requires complete audit history for wallet top-ups, payments, refunds, transfers, adjustments, expirations, reversals and administrative actions.

Record:

- Operation ID
- Wallet ID
- Customer
- Original transaction
- Operation type
- Amount

- Credit bucket
- Before balance
- After balance
- Administrator
- Approval
- Reason
- Device/IP where applicable
- Date/time
- Related transaction
- Risk result

Provide:

- Simulate
- Save Draft
- Submit for Approval
- Approve
- Reject
- Execute
- Export Audit
- View Ledger

Acceptance Condition:

Sensitive wallet operations can be validated before execution, and every financial or administrative action is permanently traceable through the wallet ledger and audit history.

Board 7 — Operational Architecture

The backend flow should be:

- Operation Request
- → Identify Wallet & Transaction
- → Permission Check
- → Eligibility Rules
- → Limits / Risk
- → Approval if Required
- → Simulation / Impact Calculation
- → Execute Ledger Entries
- → Reconciliation Event
- → Customer Notification
- → Immutable Audit

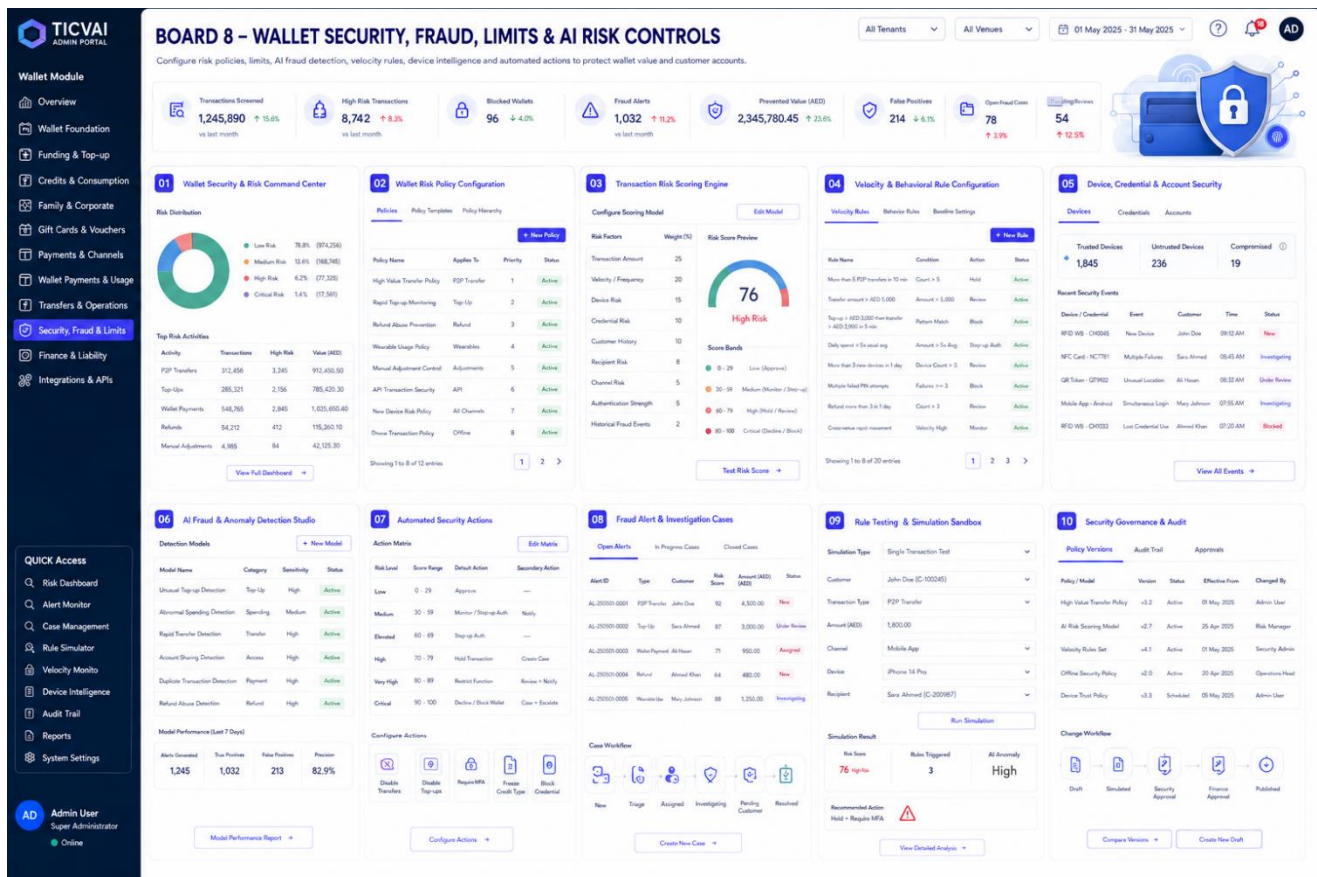
This preserves the separation across the module:

Board 1 — Wallet Foundation**Board 2 — Funding & Top-Up****Board 3 — Stored Value & Consumption****Board 4 — Family & Corporate Sharing****Board 5 — Gift Cards, Vouchers & Benefits****Board 6 — Payments, Wearables & Channels****Board 7 — Transfers, Refunds & Wallet Operations**

For **Board 8**, we should move into **Wallet Security, Fraud, Limits & AI Risk Controls**. That should be one of the most advanced boards in the Wallet module, covering suspicious activity detection, transaction risk scoring, velocity rules, account takeover protection, device/credential risk, rule-based + AI fraud detection, case review, automated actions and security monitoring—directly covering **4.3.30, 4.3.31 and 4.3.32**

Board 8 — Wallet Security, Fraud, Limits & AI Risk Controls

10 Backend Configuration Screens



Board Objective: Configure TICVAI's centralized wallet security and fraud-control layer, combining **rule-based controls, transaction risk scoring, velocity monitoring, device and credential intelligence, AI anomaly detection, automated protective actions, fraud investigation, and security governance.**

This board directly covers requirements **4.3.30, 4.3.31 and 4.3.32**, which require configurable wallet limits, suspicious-transaction controls, manual review, spending/transfer restrictions and AI identification of unusual top-ups, spending, account sharing, rapid transfers, duplicate transactions and unauthorized-access patterns.

The architecture should use:

- Rules Engine + Real-Time Risk Score + AI Behavioral Detection

rather than relying on AI alone.

Screen 01 — Wallet Security & Risk Command Center

Purpose: Provide Security, Risk, Finance and authorized Operations teams with a real-time overview of wallet security.

- Dashboard KPIs

Display:

- Transactions Screened
- Transactions Approved
- Transactions Challenged
- Transactions Declined
- Transactions Held
- High-Risk Transactions
- Wallets Under Review
- Blocked Wallets
- Suspicious Transfers

- Suspicious Top-Ups
- Account-Takeover Alerts
- Credential Security Alerts
- Open Fraud Cases
- Prevented Value
- **Risk Distribution**

Display transactions as:

- Low Risk
- Medium Risk
- High Risk
- Critical
- **Risk by Activity**

Break down:

- Top-Up
- Wallet Payment
- P2P Transfer
- Refund
- Gift Card
- Voucher
- Manual Adjustment
- Wearable Transaction
- API Transaction
- Offline Transaction

Filters

- Tenant
- Venue
- Wallet type
- Customer
- Risk level
- Channel
- Device
- Credential
- Transaction type
- Date/time

AI Capability:

AI continuously surfaces unusual behavioral patterns and prioritizes alerts according to estimated financial and operational impact.

Acceptance Condition:

Authorized users can monitor wallet security across tenants and trace each alert to its wallet, customer, credential, transaction and risk decision.

Screen 02 — Wallet Risk Policy Configuration

Purpose: Create reusable risk policies controlling wallet activities.

- **Risk Policy Types**

Configure policies for:

- Wallet Funding
- Wallet Spending
- P2P Transfers

- Refunds
- Gift Cards
- Vouchers
- Wearables
- Offline Transactions
- Administrative Adjustments
- API Transactions
- **Risk Conditions**

Rules can evaluate:

- Transaction amount
- Wallet balance
- Customer verification
- Wallet age
- Transaction frequency
- Funding source
- Device
- IP/location indicators where legally and operationally appropriate
- Channel
- Credential
- Venue
- Credit type
- Previous fraud events
- Failed authentication
- Risk score
- **Rule Actions**
- Allow
- Allow & Monitor
- Require Confirmation
- Require PIN
- Require MFA
- Hold
- Manual Review
- Restrict Wallet
- Decline
- Block
- **Rule Hierarchy**
- **Global → Tenant → Venue → Wallet Type → Transaction Type**

Acceptance Condition:

Security teams can create and publish wallet risk policies without modifying application code.

Screen 03 — Transaction Risk Scoring Engine

Purpose: Calculate a real-time risk score for sensitive wallet operations.

- **Example Risk Score**

Transaction:

- **AED 1,800 P2P Transfer**

Risk factors:

- New Device → +15
- Unusual Amount → +20

- High Velocity → +25
- New Recipient → +10
- Recent Password Reset → +20
- **Risk Score = 90 / 100**
- **Risk Bands**

Example:

- **0–29**
Low
- **30–59**
Medium
- **60–79**
High
- **80–100**
Critical
- **Configure Weighted Factors**
- Transaction amount
- Transaction frequency
- Wallet age
- Customer history
- Device trust
- Credential trust
- Recipient history
- Funding history
- Authentication
- Failed attempts
- Refund history
- Transfer history
- Channel behavior
- AI anomaly score
- **Actions by Score**
- Low → Approve
- Medium → Monitor / Step-up authentication
- High → MFA / Hold
- Critical → Decline + Review

Acceptance Condition:

Each configured wallet operation receives a reproducible risk score with traceable contributing factors and an associated decision.

Screen 04 — Velocity & Behavioral Rule Configuration

Purpose: Detect abnormal activity based on frequency, amount and behavioral patterns.

Requirement 4.3.31 requires controls for suspicious transactions, spending restrictions, transfer restrictions and high-risk activity.

- **Velocity Rules**

Configure:

- Transactions per minute
- Transactions per hour
- Transactions per day
- Value per hour
- Value per day
- Top-ups per hour

- Transfers per hour
- Refunds per day
- Failed attempts
- Different devices used
- Different credentials used
- Number of recipients
- Number of venues
- **Example**

Rule:

- More than 5 P2P transfers within 10 minutes

Action:

- **Hold additional transfers + Risk Review**

Another rule:

- AED 3,000 top-up followed by AED 2,900 transfer within 2 minutes

Action:

- **High-Risk Alert**
- **Behavioral Baseline**

Compare activity against:

- Customer historical behavior
- Wallet-type norms
- Membership segment
- Venue averages
- Transaction category norms

Acceptance Condition:

TICVAI evaluates configurable velocity windows and behavioral deviations in real time before authorizing applicable operations.

Screen 05 — Device, Credential & Account Security

Purpose: Detect compromised accounts, shared credentials and suspicious device behavior.

Requirement 4.3.32 specifically includes detection of **account sharing and unauthorized access patterns**.

- **Device Intelligence**

Track:

- Device ID
- First seen
- Last seen
- Customer association
- Wallet associations
- Trusted/untrusted status
- Failed authentications
- Credential changes
- Risk history
- **Credential Monitoring**

Support:

- RFID Wristband
- NFC
- QR
- Mobile Credential

- Digital Key
- Customer Login
- API Credential
- **Security Rules**

Detect:

- Same credential used simultaneously
- Same wallet on excessive devices
- Lost/stolen credential use
- New device + high-value transaction
- Rapid device switching
- Repeated PIN failures
- Repeated OTP failures
- Suspicious account recovery
- Credential cloning indicators
- **Administrative Actions**
- Trust Device
- Untrust Device
- Revoke Session
- Suspend Credential
- Force Reauthentication
- Reset Credential
- Block Wallet

Acceptance Condition:

TICVAI can identify and restrict suspicious account, device and credential activity without changing the customer's underlying balance.

Screen 06 — AI Fraud & Anomaly Detection Studio

Purpose: Configure AI-driven detection of suspicious wallet behavior that fixed rules may not identify.

Requirement 4.3.32 explicitly requires AI monitoring for unusual top-ups, abnormal spending, account sharing, rapid transfers, duplicate transactions and unauthorized access.

- **AI Detection Categories**

Enable/disable:

- Unusual Top-Up
- Abnormal Spending
- Rapid Transfers
- Duplicate Transactions
- Account Sharing
- Unauthorized Access
- Refund Abuse
- Gift Card Abuse
- Voucher Abuse
- Credential Misuse
- Unusual Offline Activity
- Manual Adjustment Anomaly
- **AI Sensitivity**

Configure:

- **Low / Balanced / High / Custom**
- **AI Inputs**

Subject to configured privacy/security policy:

- Historical transaction patterns
- Typical transaction amount
- Typical venue/channel
- Usual device
- Frequency
- Time patterns
- Credit usage
- Recipient relationships
- Refund behavior
- **Explainability**

Every AI alert should show:

- **Why was this flagged?**

Example:

Transaction amount is 6.4× higher than this wallet's normal spending.

Three new recipients were added within 15 minutes.

Device has never previously accessed this wallet.

AI should provide **risk evidence**, not an unexplained black-box decision.

Acceptance Condition:

AI-generated alerts contain understandable contributing factors and are governed by configurable thresholds and human-review policies.

Screen 07 — Automated Security Action Orchestration

Purpose: Determine what TICVAI automatically does when a security or fraud condition occurs.

- **Available Actions**
- **Low Risk**
→ Approve
- **Medium Risk**
→ Approve + Monitor
- **Elevated Risk**
→ Step-Up Authentication
- **High Risk**
→ Hold Transaction
- **Very High Risk**
→ Restrict Function
- **Critical**
→ Block Wallet + Create Case
- **Granular Restrictions**

Instead of always blocking the whole wallet, TICVAI can:

- Block P2P transfers
- Disable top-ups
- Disable online payments
- Disable wearable payments
- Disable specific credential
- Freeze specific credit type
- Require MFA
- Set temporary spending limit
- Allow balance inquiry only

Configure

- Trigger
- Risk score
- Action
- Duration
- Customer notification
- Security notification
- Case creation
- Approval for release
- Automatic expiry
- Escalation
- **Example**
- **Suspected credential compromise**

TICVAI:

1. Suspends credential.
2. Keeps wallet active.
3. Requires reauthentication.
4. Notifies customer.
5. Creates security case.

This avoids unnecessarily blocking legitimate wallet value.

Acceptance Condition:

Configured risk events trigger deterministic and auditable protective actions appropriate to their severity.

Screen 08 — Fraud Alert & Investigation Case Management

Purpose: Provide security teams with a structured investigation workspace.

- **Alert Queue**

Display:

- Alert ID
- Wallet
- Customer
- Transaction
- Amount
- Risk score
- AI score
- Triggered rules
- Device
- Credential
- Venue
- Date/time
- Priority
- **Status**
- **Case Workflow**
- **New**
- → **Triage**
- → **Under Investigation**
- → **Customer Verification**
- → **Escalated**
- → **Confirmed Fraud / False Positive**

- → **Resolved**
- → **Closed**
- **Investigator Workspace**

Show:

- Wallet timeline
- Transaction history
- Funding history
- Transfer network
- Devices
- Credentials
- Authentication history
- Refund history
- Related alerts
- Previous cases
- **Investigator Actions**
- Mark Safe
- Hold Transaction
- Block Transfer
- Suspend Credential
- Freeze Wallet
- Request Verification
- Reverse where permitted
- Escalate
- Close as False Positive
- Confirm Fraud

AI Capability:

AI summarizes the suspicious activity and groups related alerts to reduce investigation time.

Acceptance Condition:

Every security alert can be investigated through a controlled case workflow with complete decision history.

Screen 09 — Security Rules Testing, Simulation & AI Sandbox

Purpose: Test new fraud rules and AI configurations before applying them to live transactions.

This screen is particularly important because overly aggressive security rules can create large numbers of false declines.

- **Simulation Modes**
- **Single Transaction Test**

Enter:

- Wallet
- Customer
- Amount
- Channel
- Device
- Credential
- Transaction type

TICVAI returns:

- Risk Score → **74**
- Rules Triggered → **3**
- AI Anomaly → **High**
- Decision → **MFA + Hold**

- **Historical Replay**

Administrators can safely test a proposed rule against historical transactions.

Example:

- “What would have happened if this rule had been active during the previous 30 days?”

Display:

- Transactions Evaluated → 1,245,000
- Would Approve → 98.4%
- Would Challenge → 1.1%
- Would Hold → 0.4%
- Would Decline → 0.1%
- **Impact Analysis**

Show:

- Potential fraud detected
- Potential false positives
- Customer impact
- Financial exposure
- Channels affected
- Wallet types affected

AI Capability:

AI recommends threshold adjustments when simulations indicate excessive false positives or weak detection.

Acceptance Condition:

New security policies can be tested against simulated or appropriately controlled historical scenarios before production publication.

Screen 10 — Security Governance, Audit & Rule Publication

Purpose: Govern how wallet security policies, fraud models and automated actions are changed and released.

- **Configuration Areas**

Display:

- Risk policies
- Rule versions
- AI configuration version
- Thresholds
- Automated actions
- Effective dates
- Applicable tenants
- Applicable venues
- Owner
- Approval status
- **Change Governance**

Every change records:

- Old configuration
- New configuration
- User
- Date/time
- Reason
- Approval
- Effective date
- **Publication Workflow**

- **Draft**
- → **Simulation Required**
- → **Validation**
- → **Security Approval**
- → **Finance/Operations Approval where required**
- → **Scheduled**
- → **Published**
- → **Monitored**

Support:

- Version comparison
- Rollback
- Emergency disable
- Rule suspension
- Effective dating
- **Security Audit**

Maintain history for:

- Risk decisions
- Fraud alerts
- Blocks
- Restrictions
- MFA challenges
- Credential suspension
- Manual overrides
- Case decisions
- Rule changes
- AI configuration changes

Acceptance Condition:

No material wallet-security configuration can enter production without appropriate validation, permissions, versioning and approval, and all changes remain permanently auditable.

Board 8 — Real-Time Risk Decision Flow

We would specify the backend sequence clearly:

- **Wallet Transaction Request**
- → **Identity / Credential Validation**
- → **Static Security Rules**
- → **Velocity Check**
- → **Device & Credential Risk**
- → **Customer Behavioural Profile**
- → **AI Anomaly Detection**
- → **Combined Risk Score**
- → **Decision Engine**

Then:

- **LOW**
→ Approve
- **MEDIUM**
→ Approve + Monitor
- **ELEVATED**
→ PIN / MFA / Digital ID
- **HIGH**
→ Hold + Review
- **CRITICAL**
→ Decline / Restrict + Create Fraud Case
- → **Immutable Risk & Decision Audit**

This makes Board 8 significantly more sophisticated than simply creating a few fraud rules. It becomes a genuine **TICVAI Wallet Risk Engine**.

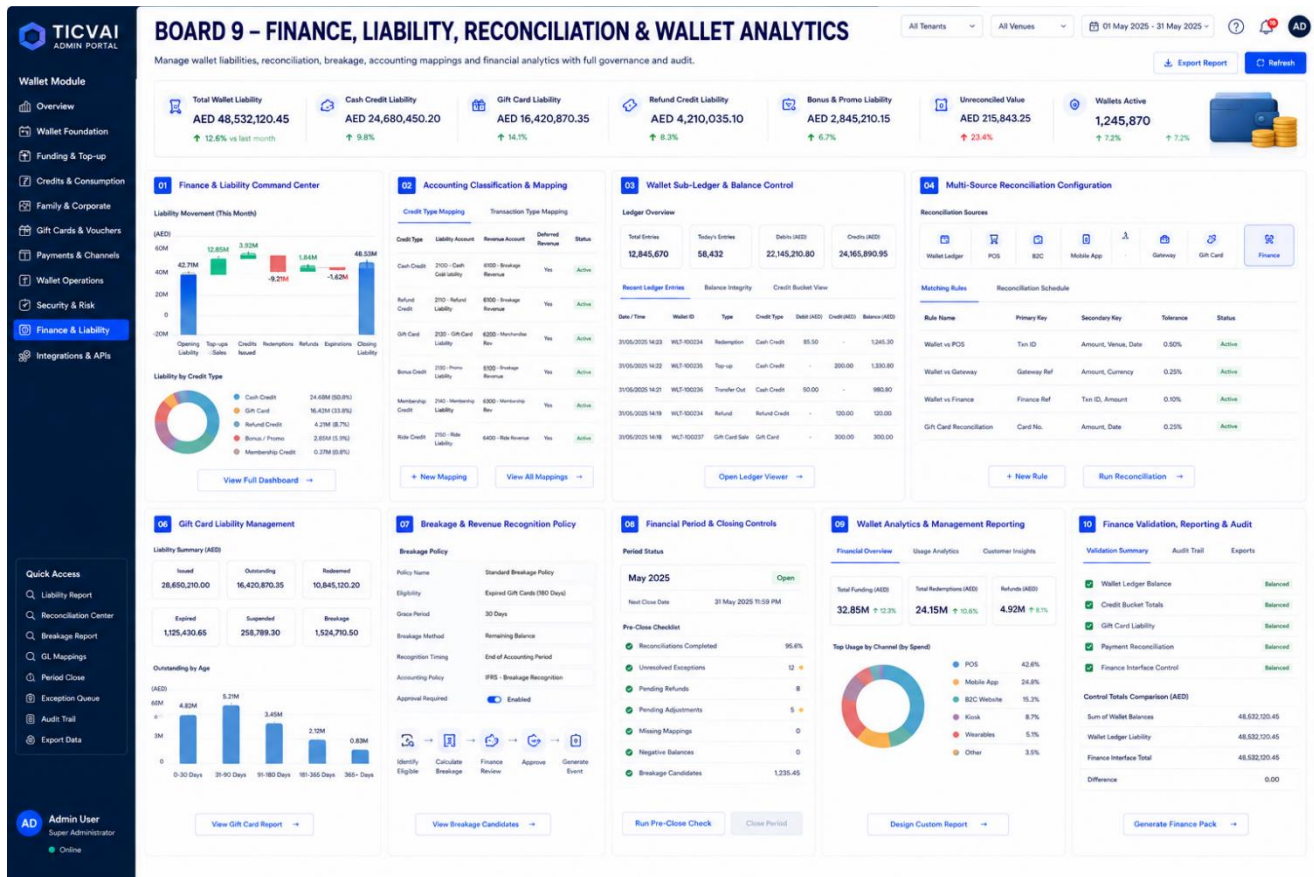
The module progression is now:

Board 1 — Wallet Foundation**Board 2 — Funding & Top-Up****Board 3 — Stored Value & Consumption****Board 4 — Family & Corporate Wallets****Board 5 — Gift Cards, Vouchers & Benefits****Board 6 — Payments, Wearables & Channels****Board 7 — Transfers, Refunds & Operations****Board 8 — Security, Fraud & AI Risk Controls**

For **Board 9**, I recommend **Finance, Liability, Reconciliation & Wallet Analytics**. This is where we should bring together wallet liability, gift-card liability, breakage, revenue recognition events, transaction reconciliation, settlement controls, finance mappings, wallet analytics and management reporting—particularly requirements **4.3.17**, **4.3.33**, **4.3.36** and **4.3.37**, while keeping the actual general ledger/accounting engine in the Finance module.

Board 9 — Finance, Liability, Reconciliation & Wallet Analytics

10 Backend Configuration Screens



Board Objective: Configure the financial-control layer of TICVAI Wallet, providing complete visibility and governance over wallet liabilities, stored-value balances, gift-card exposure, reconciliation, breakage, revenue-recognition events, accounting mappings, financial exceptions and management analytics.

This board directly addresses requirements 4.3.17, 4.3.33, 4.3.36 and 4.3.37, as well as the reconciliation requirement described under 4.3.4.

The architectural boundary should remain clear:

Wallet Module = sub-ledger and source of wallet financial events.

Finance Module = accounting/GL, financial posting and corporate financial reporting.

Board 9 therefore controls the wallet-side financial configuration and sends governed accounting events to the TICVAI Finance module.

Screen 01 — Wallet Finance & Liability Command Center

Purpose: Provide Finance and authorized management users with a consolidated financial view of all wallet obligations and movements.

- Executive KPIs

Display:

- Total Wallet Liability
- Cash Credit Liability
- Gift Card Liability
- Refund Credit Liability
- Bonus/Promotional Value
- Membership Credit Exposure
- Outstanding Stored Value

- Issued Value
- Redeemed Value
- Expired Value
- Breakage Recognized
- Unreconciled Value
- Suspended/Frozen Value
- **Liability Breakdown**

By:

- Credit type
- Wallet type
- Gift card
- Tenant
- Venue
- Currency
- Customer type
- Accounting period
- **Movement Analysis**

Show:

- **Opening Liability**
- Funding
- Gift Cards Issued
- Credits Issued
- Refunds to Wallet
 - Redemptions
 - Expirations
 - Reversals/Corrections
- **= Closing Liability**
- **Alerts**

Highlight:

- Liability spikes
- Unusual breakage
- Unreconciled transactions
- Negative balances
- Accounting mapping failures
- Large manual adjustments
- Expired value awaiting treatment

AI Capability: AI explains significant liability movements and identifies unusual changes requiring finance review.

Acceptance Condition: Authorized finance users can reconcile the consolidated wallet liability back to individual wallet balances and ledger transactions.

Screen 02 — Wallet Financial Classification & Accounting Mapping

Purpose: Define the financial classification of every wallet credit and transaction type.

- **Credit Mapping**

Configure accounting treatment for:

- Cash Credit
- Refund Credit
- Gift Card
- Bonus Credit

- Promotional Credit
- Membership Credit
- Ride Credit
- F&B Credit
- Retail Credit
- Parking Credit
- Other credits
- **Transaction Mapping**

Configure:

- Top-Up
- Redemption
- Refund
- Transfer
- Adjustment
- Reversal
- Expiration
- Gift Card Sale
- Gift Card Redemption
- Breakage
- **Mapping Dimensions**

Each mapping may include:

- Wallet transaction type
- Credit type
- Liability classification
- Revenue classification
- Deferred revenue category
- GL account reference
- Cost center
- Business unit
- Venue
- Department
- Currency
- Tax treatment
- Effective date
- **Example**
- **Gift Card Sale**
- Cash/Payment Clearing
→ Gift Card Liability
- **Gift Card Redemption**
- Gift Card Liability
→ Applicable Revenue Event

The final accounting entry remains the responsibility of TICVAI Finance.

Acceptance Condition: Every financially relevant wallet event has an effective-dated accounting classification before it can be sent downstream for financial posting.

Screen 03 — Wallet Sub-Ledger & Balance Control

Purpose: Maintain the authoritative financial transaction history behind every wallet balance.

- **Sub-Ledger View**

Each entry records:

- Ledger Entry ID
- Wallet ID
- Customer/account
- Transaction ID
- Transaction type
- Credit bucket
- Credit lot
- Debit
- Credit
- Currency
- Balance before
- Balance after
- Source system
- Venue
- Channel
- Timestamp
- Financial status
- Related transaction
- **Ledger Principles**

The ledger should be:

- Immutable
- Chronological
- Double-reference controlled
- Version protected
- Fully auditable

No financial transaction should be deleted.

Corrections occur through:

- **Original Entry**
- → **Reversal Entry**
- → **Corrected Entry**
- **Balance Integrity Check**

TICVAI continuously verifies:

Sum of ledger entries = Wallet balance.

Any mismatch generates a financial exception.

Acceptance Condition: Every displayed wallet balance can be reconstructed from its immutable sub-ledger history.

Screen 04 — Multi-Source Reconciliation Configuration

Purpose: Reconcile wallet financial events across Wallet, Payments, Sales Channels and Finance.

The supplied requirements specifically call for reconciliation of payment status between sales channels, the bank/payment gateway and the ticketing system.

- **Reconciliation Sources**

Support:

- Wallet Ledger
- POS
- B2C
- Mobile App

- Kiosk
- Payment Gateway
- Payment Server
- Gift Card
- F&B
- Retail
- Ticketing
- Finance
- Third-party integrations
- **Matching Keys**

Configure:

- Transaction ID
- Wallet ID
- Payment reference
- Order number
- Amount
- Currency
- Date
- Venue
- Terminal
- External reference
- **Reconciliation Status**
- **Matched**
- **Partially Matched**
- **Unmatched**
- **Duplicate**
- **Amount Mismatch**
- **Currency Mismatch**
- **Missing Wallet Entry**
- **Missing Payment Entry**
- **Reconciliation Frequency**
- Real-time
- Hourly
- End of day
- Scheduled
- Manual

Acceptance Condition: TICVAI can systematically compare wallet events against configured source systems and identify financial discrepancies.

Screen 05 — Reconciliation Exception & Resolution Workbench

Purpose: Provide Finance and Operations with a governed workspace to investigate reconciliation failures.

- **Exception Examples**
- **Payment successful / Wallet not funded**
- **Wallet funded / Payment failed**
- **Wallet debited / POS transaction missing**
- **Duplicate wallet debit**
- **Refund issued / Finance event missing**
- **Gift card redeemed / Liability unchanged**

Exception Queue

Display:

- Exception ID
- Transaction
- Wallet
- Source
- Expected amount
- Actual amount
- Difference
- Currency
- Age
- Priority
- Owner
- **Status**

Workflow

- **New**
- → **Assigned**
- → **Investigating**
- → **Pending External System**
- → **Correction Proposed**
- → **Approved**
- → **Resolved**
- **Resolution Actions**
- Retry synchronization
- Create compensating transaction
- Reverse
- Correct mapping
- Escalate
- Mark justified
- Send to Finance
- Send to Payments

AI Capability

- AI groups related exceptions and suggests likely root causes such as:

Payment gateway synchronization delay.

Duplicate POS retry.

Missing wallet callback.

Incorrect accounting mapping.

Acceptance Condition: Every reconciliation discrepancy is managed through a traceable resolution workflow without deleting or silently modifying financial records.

Screen 06 — Gift Card Liability Management

Purpose: Provide detailed financial control of outstanding gift-card obligations.

Requirement **4.3.36** explicitly requires reporting of outstanding gift-card balances, redeemed value, unredeemed balances, expired balances and liability exposure by venue, tenant, currency and accounting period.

- **Liability Dashboard**

Display:

- Gift Cards Issued

- Original Issued Value
- Outstanding Value
- Redeemed Value
- Partially Redeemed Value
- Unredeemed Value
- Expired Value
- Suspended Value
- Breakage
- Liability Exposure
- **Breakdown**

By:

- Gift card program
- Tenant
- Venue
- Currency
- Issuance date
- Expiry date
- Accounting period
- Sales channel
- Corporate program
- **Aging Analysis**

For example:

Age	Outstanding
0–30 Days	AED 2.5M
31–90 Days	AED 1.8M
91–180 Days	AED 1.1M
181–365 Days	AED 750K
365+ Days	AED 420K

- **Drill Down**
- **Program → Gift Card → Customer → Transactions**

Acceptance Condition: Finance can trace reported gift-card liability to individual active and historical gift-card balances.

Screen 07 — Breakage & Revenue Recognition Policy

Purpose: Configure the wallet-side business rules for expired/unredeemed gift-card value.

Requirement **4.3.37** requires configurable expiration policies, gift-card breakage calculation and generation of revenue-recognition entries according to configured accounting policies.

- **Breakage Configuration**

Configure:

- Breakage enabled
- Eligible gift-card programs
- Expiry requirement
- Grace period
- Recognition method
- Recognition timing
- Accounting policy reference
- Jurisdiction/business-rule profile
- Approval requirement

- **Supported Treatment**

Depending on configured accounting policy:

- Maintain liability
- Recognize on expiry
- Recognize after grace period
- Recognize according to configured breakage calculation
- Manual finance review
- **Example**

Gift Card:

- Original value → AED 500
- Redeemed → AED 350
- Expired remaining balance → AED 150

Configured treatment:

- **AED 150 → Breakage Candidate**

Finance approval:

- **Approved**

→ Generate Finance Recognition Event.

- **Governance**

Wallet does **not** independently decide accounting compliance.

Instead:

- **Wallet identifies eligible value**
- → **Policy calculates proposed treatment**
- → **Finance approval / Finance Module**
- → **Accounting event**

Acceptance Condition: Breakage candidates are calculated according to configured policy and no recognition event is finalized without applicable financial governance.

Screen 08 — Wallet Financial Period & Closing Controls

Purpose: Support controlled month-end and financial-period processing for wallet balances.

- **Period Status**
- **Open**
- → **Closing**
- → **Under Review**
- → **Closed**
- → **Reopened with Authorization**
- **Pre-Close Validation**

TICVAI checks:

- Unreconciled transactions
- Pending refunds
- Pending reversals
- Pending adjustments
- Negative balances
- Missing accounting mappings
- Failed integrations
- Unprocessed expirations
- Gift-card breakage candidates
- Pending approvals

- **Closing Snapshot**

Capture:

- Opening liability
- Funding
- Credits issued
- Redemption
- Refunds
- Transfers
- Adjustments
- Expiration
- Breakage
- Closing liability
- **Controls**
- Close by tenant
- Close by venue
- Close by currency
- Lock financial period
- Reopen with approval
- Carry exceptions forward
- Export close package

Acceptance Condition: A financial period cannot close until configured critical validation checks have passed or exceptions have been formally approved.

Screen 09 — Wallet Analytics & Management Reporting

Purpose: Provide comprehensive analytics for wallet usage, financial performance and customer behavior.

Requirement **4.3.17** requires intensive reporting across wallet credit types, including usage, balance and expiry, while **4.3.33** calls for dashboards covering balances, top-ups, redemptions, refunds, outstanding liability, expired value, usage by channel and transaction volume.

- **Financial Analytics**
- Wallet Liability
- Outstanding Stored Value
- Funding
- Redemption
- Refunds
- Transfers
- Expired Value
- Breakage
- Gift Card Liability
- **Operational Analytics**
- Active wallets
- Average wallet balance
- Average top-up
- Average spend
- Transaction volume
- Wallet usage frequency
- Dormant wallets
- Credit utilization
- **Channel Analytics**

Compare:

- POS
- Mobile App
- B2C
- Kiosk
- Wearables
- API
- F&B
- Retail
- Attractions
- **Customer Analytics**

Analyze:

- Individual
- Family
- Membership
- Corporate
- Employee
- Guest
- **AI Insights**

Example:

Wallet-funded customers spend 18% more per venue visit.

AED 1.2M of promotional credits expire within 30 days.

Gift-card redemption increased 14% after mobile-wallet integration.

Venue A has unusually high refund-credit liability.

Allow reports to be:

- Saved
- Scheduled
- Exported
- Shared according to permissions

Acceptance Condition: Authorized users can analyze wallet financial and operational performance from aggregate KPI level down to the supporting transaction data.

Screen 10 — Finance Validation, Reporting & Audit Center

Purpose: Provide a final finance-control screen for validating wallet financial integrity and reviewing all financial configuration changes.

- **Finance Health Check**

Validate:

- **Wallet Ledger Balance**
→ PASS
- **Credit Bucket Totals**
→ PASS
- **Gift Card Liability**
→ PASS
- **Payment Reconciliation**
→ PASS
- **Finance Mapping**
→ PASS
- **Breakage Processing**
→ PASS

- **Financial Period**
→ PASS
- **Control Totals**

Compare:

- Sum of Customer Wallet Balances
- against
- Wallet Sub-Ledger Liability
- against
- Finance Interface Control Total

Any difference becomes a visible exception.

- **Configuration Audit**

Record changes to:

- Accounting mappings
- Liability policies
- Breakage rules
- Reconciliation rules
- Financial periods
- Manual corrections
- Financial approvals

For every change record:

- User
- Role
- Old value
- New value
- Reason
- Approval
- Date/time
- Effective date

Actions

- Run Finance Validation
- Review Exceptions
- Submit Close
- Approve Close
- Generate Liability Report
- Generate Reconciliation Report
- Export Audit
- Send to Finance

AI Capability: AI summarizes material discrepancies and identifies the transactions most likely responsible for control-total differences.

Acceptance Condition: Finance users can prove wallet financial integrity from reported balances through the wallet sub-ledger and downstream finance interface, with all financial configuration and corrective actions fully auditable.

Board 9 — Financial Architecture

We would make this separation explicit:

- Customer Wallets
- ↓
- Credit Buckets & Gift Cards
- ↓
- Immutable Wallet Sub-Ledger
- ↓
- Wallet Liability Engine
- ↓
- Reconciliation Engine
- ↙ Payments / Gateway
- ↙ POS / B2C / App / F&B / Retail
- ↓
- Financial Control & Period Close
- ↓
- Finance Integration Events
- ↓
- TICVAI Finance Module / GL

This prevents us from accidentally building a second accounting system inside Wallet.

The **Wallet module owns the financial truth of the wallet**, while the **Finance module owns the accounting truth of the company**.

So we now have:

Board 1 — Wallet Foundation

Board 2 — Funding & Top-Up

Board 3 — Stored Value & Consumption

Board 4 — Family & Corporate Wallets

Board 5 — Gift Cards, Vouchers & Benefits

Board 6 — Payments, Wearables & Channels

Board 7 — Transfers, Refunds & Operations

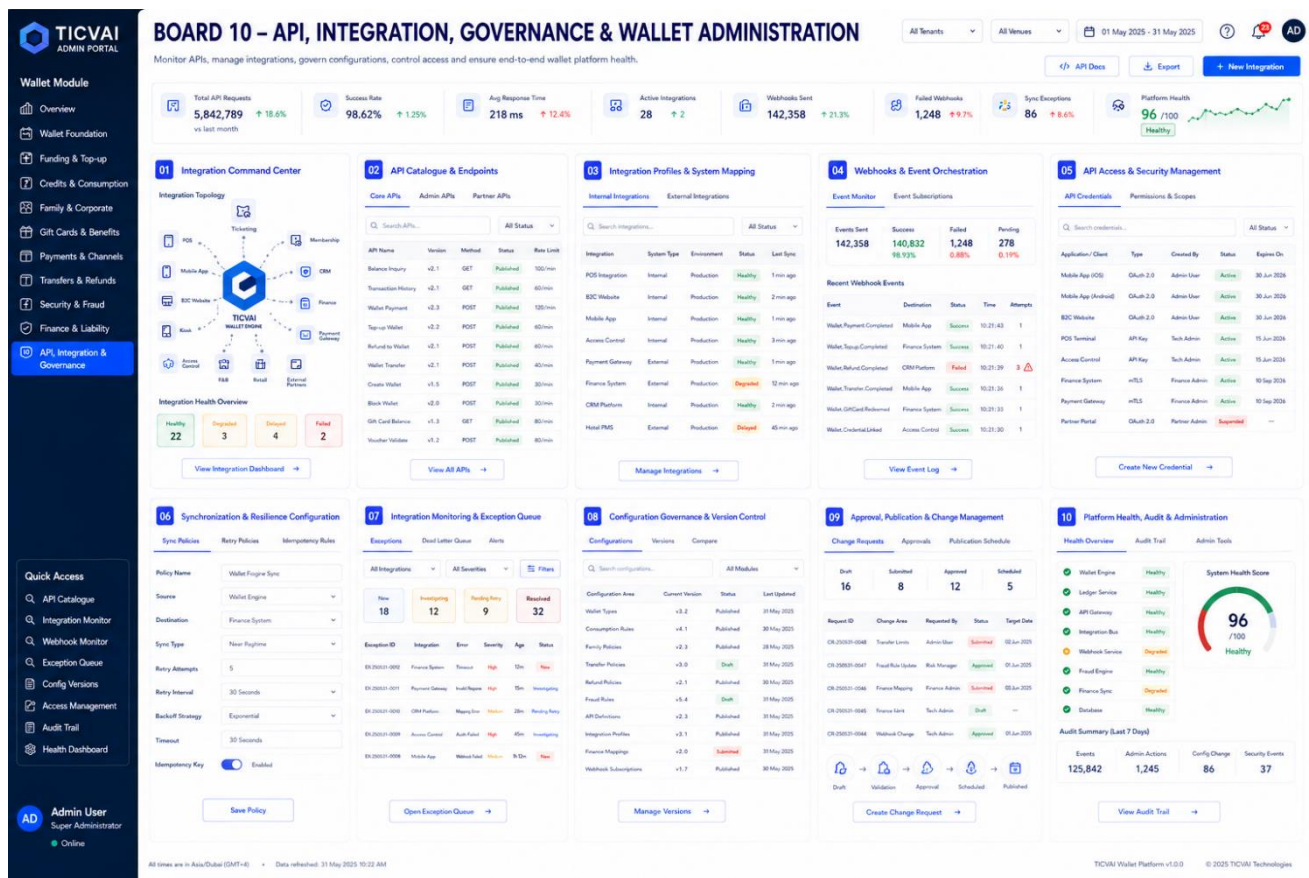
Board 8 — Security, Fraud & AI Risk

Board 9 — Finance, Liability, Reconciliation & Analytics

That leaves **Board 10 as the final board: API, Integration, Governance & Wallet Administration**. It should close the entire module by covering wallet APIs, external-system integrations, webhooks/events, API permissions, synchronization, integration monitoring, configuration versioning, approval/publication, audit governance and end-to-end wallet health. This will directly cover **4.3.20 and 4.3.34** and complete the architecture without duplicating the previous nine boards.

Board 10 — API, Integration, Governance & Wallet Administration

- 10 Backend Configuration Screens — Final Wallet Board



Board Objective: Complete the TICVAI Wallet module with the enterprise integration and governance layer that allows Wallet to operate securely across the entire TICVAI ecosystem and with approved third-party platforms.

This board covers **Wallet APIs, integration profiles, event/webhook orchestration, synchronization, integration monitoring, access/security governance, configuration versioning, approval and publication, audit governance, and end-to-end platform health.**

It directly addresses requirements **4.3.20** and **4.3.34**, while consolidating governance and administration capabilities required across the full wallet scope. The source specifically requires integration with internal and external systems through APIs, including balance inquiry, transaction history, wallet funding, wallet payment, refund processing and wallet-to-wallet transfers.

Screen 01 — Wallet Integration Command Center

Purpose: Provide administrators and technical operations teams with a centralized view of all Wallet integrations and API activity.

Integration KPIs

Display:

- Active Integrations
- Connected TICVAI Modules
- External Integrations
- API Requests Today
- Successful API Requests
- Failed API Requests
- API Success Rate

- Average Response Time
- Webhook Events
- Failed Webhooks
- Synchronization Failures
- Integration Alerts
- **Connected TICVAI Domains**

Display connectivity with:

- Ticketing
- Event Management
- Membership
- CRM
- Pricing
- Promotions
- POS
- F&B
- Retail
- Rental
- Parking
- Access Control
- Finance
- Payments
- Reporting
- **Integration Status**

Each integration displays:

- **Healthy**
- **Degraded**
- **Delayed**
- **Failed**
- **Suspended**
- **Maintenance**

Filters

- Tenant
- Venue
- Integration
- API
- Environment
- **Status**
- Transaction type
- Date/time

Quick Actions

- Create Integration
- View API Health
- Retry Failed Event
- Open Error Log
- Suspend Integration
- Test Connection

AI Capability:

AI detects integration degradation and correlates failures across services to suggest likely root causes.

Acceptance Condition:

Authorized administrators can monitor Wallet connectivity and identify failed, degraded or delayed integrations from a centralized workspace.

Screen 02 — Wallet API Catalogue & Endpoint Configuration

Purpose: Define and govern the API services exposed by the TICVAI Wallet Engine.

Requirement **4.3.34** explicitly requires APIs supporting wallet operations.

- **Core Wallet APIs**

Configure APIs for:

- **Wallet Management**
 - Create Wallet
 - Retrieve Wallet
 - Update Wallet
 - Block Wallet
 - Unblock Wallet
- **Balance**
 - Balance Inquiry
 - Balance by Credit Type
 - Available Balance
 - Reserved Balance
- **Transactions**
 - Transaction History
 - Transaction Detail
 - Wallet Payment
 - Wallet Redemption
- **Funding**
 - Fund Wallet
 - Top-Up
 - Auto-Reload
- **Transfers**
 - Wallet-to-Wallet Transfer
 - Transfer Status
- **Refunds**
 - Refund to Wallet
 - Refund Status
- **Gift Cards / Benefits**
 - Gift Card Balance
 - Voucher Validation
 - Benefit Inquiry
- **Endpoint Configuration**

For each API define:

- API name
- Endpoint reference
- Version
- Method
- Request schema
- Response schema
- Authentication
- Permission scope

- Tenant scope
- Rate limit
- Timeout
- Idempotency requirement
- Logging level
- Environment
- **Status**

API Lifecycle

- **Draft → Test → Approved → Published → Deprecated → Retired**

Acceptance Condition:

Every wallet capability exposed externally is registered, secured, versioned and governed through the Wallet API catalogue.

Screen 03 — Integration Profile & System Mapping

Purpose: Configure how TICVAI Wallet communicates with internal modules and approved external systems.

Requirement **4.3.20** requires wallet integration with both internal and external systems through APIs.

Integration Types

- TICVAI Internal Service
- Payment Platform
- POS System
- Mobile Application
- B2C Website
- Hotel/PMS
- Access Control
- CRM
- Loyalty Platform
- Finance/ERP
- Partner Platform
- Third-party Venue System
- External Gift Card Platform
- **Integration Profile**

Configure:

- Integration name
- System type
- Owner
- Environment
- Base endpoint
- Authentication profile
- Supported operations
- Tenant
- Venue
- Currency
- Timeout
- Retry policy
- Error handling
- Data mapping
- Effective dates

Data Mapping

Example:

- **External Wallet ID**
↔ TICVAI Wallet ID
- **External Customer ID**
↔ TICVAI Customer ID
- **External Transaction Code**
↔ TICVAI Transaction Type
- **Mapping Transformation**

Support:

- Field mapping
- Code translation
- Currency mapping
- Status mapping
- Credit-type mapping
- Date/time mapping

Acceptance Condition:

Each connected platform has an explicit integration profile and controlled mapping to TICVAI wallet objects.

Screen 04 — Wallet Events, Webhooks & Notification Orchestration

Purpose: Configure event-driven communication when something changes in Wallet.

Wallet Events

Support events such as:

- Wallet Created
- Wallet Activated
- Wallet Blocked
- Wallet Unblocked
- Balance Changed
- Top-Up Completed
- Top-Up Failed
- Payment Completed
- Payment Declined
- Refund Completed
- Transfer Completed
- Transfer Failed
- Credit Issued
- Credit Expiring
- Credit Expired
- Gift Card Activated
- Gift Card Redeemed
- Credential Linked
- Credential Suspended
- Fraud Alert Created
- **Webhook Configuration**

For each subscriber configure:

- Event
- Destination
- Authentication

- Payload version
- Tenant
- Venue
- Retry count
- Retry interval
- Timeout
- Signing/security
- Failure behavior

Delivery Status

- **Queued**
- → **Sent**
- → **Acknowledged**
- or
- **Failed → Retry → Dead Letter**

Event Replay

Authorized administrators can replay eligible failed events without recreating the original wallet transaction.

Acceptance Condition:

Wallet events are reliably distributed to authorized subscribing systems with complete delivery and retry history.

Screen 05 — API Security, Access & Integration Permissions

Purpose: Ensure integrations receive only the wallet permissions and data required for their function.

API Permission Scopes

Examples:

- wallet.read
- wallet.create
- balance.read
- transaction.read
- payment.execute
- topup.execute
- refund.execute
- transfer.execute
- giftcard.read
- voucher.redeem
- credential.manage

Configure

- Integration identity
- Application/client
- Tenant scope
- Venue scope
- API scope
- Allowed operations
- Allowed wallet types
- Maximum transaction
- Environment
- IP/network restrictions where applicable
- Credential expiry
- Secret/key rotation
- MFA for administration

- Certificate configuration
- **Access Models**

Support:

- Read Only
- Transactional
- Administrative
- Finance
- Partner
- Internal Service
- **High-Risk Operations**

Operations such as:

- **Balance Adjustment**
- **Wallet Block**
- **High-Value Refund**
- **Administrative Transfer**

should require stronger permissions than simple balance inquiry.

- **Emergency Controls**
- Revoke credential
- Disable integration
- Disable API scope
- Rotate key
- Suspend transaction capability

Acceptance Condition:

Every API request is authenticated and authorized against explicit integration, tenant, venue and operation-level permissions.

Screen 06 — Synchronization, Retry & Resilience Configuration

Purpose: Configure how Wallet behaves when dependent services or external systems are temporarily unavailable.

Synchronization Types

- Real-Time
- Near Real-Time
- Batch
- Scheduled
- Event Driven
- Manual

Configure

- Retry attempts
- Retry interval
- Exponential backoff
- Timeout
- Idempotency
- Duplicate prevention
- Queue behavior
- Dead-letter queue
- Event retention
- Replay permission
- Dependency priority
- **Example**

Wallet refund completed.

Finance integration unavailable.

TICVAI should **not reverse the valid wallet refund** simply because Finance is temporarily unavailable.

Instead:

- **Wallet Refund = Completed**
- **Finance Event = Pending**
- → Retry
- → Retry

→ Escalate if unresolved.

- **Idempotency**
- Repeated API requests with the same transaction key must not create duplicate:
- Top-ups
- Payments
- Refunds
- Transfers
- Adjustments

Acceptance Condition:

Temporary integration failures do not cause duplicate financial transactions, lost wallet events or inconsistent wallet balances.

Screen 07 — Integration Monitoring & Exception Workbench

Purpose: Give technical and operational teams a dedicated workspace for failed integrations.

Exception Categories

- Authentication failure
- Timeout
- Invalid payload
- Mapping failure
- API unavailable
- Duplicate request
- Currency mismatch
- Wallet not found
- Customer mapping missing
- Transaction rejected
- Webhook failure
- Finance synchronization failure

Exception Queue

Display:

- Exception ID
- Integration
- Wallet
- Transaction
- Endpoint/event
- Error code
- Error message
- Attempt count
- First failure
- Last retry
- Priority

- Owner
- **Status**

Workflow

- **New**
- → **Assigned**
- → **Investigating**
- → **Retrying**
- → **Pending External System**
- → **Resolved**
- → **Closed**

Actions

- Retry
- Replay
- Correct mapping
- Reprocess
- Escalate
- Suspend integration
- Mark resolved

AI Capability

AI correlates failures and may identify patterns such as:

84% of current wallet failures originate from Payment Gateway X.

or:

Failures began immediately after API version 3.4 deployment.

Acceptance Condition:

Every integration failure is traceable through a controlled exception workflow without modifying the original financial transaction.

Screen 08 — Wallet Configuration Governance & Version Control

Purpose: Provide centralized governance for configuration changes across all ten Wallet boards.

This screen becomes the **master configuration governance layer** for the entire Wallet module.

Governed Configuration Areas

- Wallet Types
- Funding Rules
- Credit Types
- Consumption Rules
- Family Policies
- Corporate Policies
- Gift Cards
- Vouchers
- Channel Policies
- Wearables
- Transfer Policies
- Refund Policies
- Security Rules
- Fraud Models
- Finance Mappings
- APIs

- Integrations

Versioning

Every configuration maintains:

- Version
- **Status**
- Owner
- Created date
- Modified date
- Effective from
- Effective to
- Approved by
- Publication status

Compare Versions

Example:

- **Wallet Consumption Policy**
- Version 3.1

vs.

- Version 3.2

Show:

- **Cash Priority:** 3 → 5
- **Bonus Priority:** 2 → 1
- **FEFO:** Enabled → Enabled
- **Gift Card Priority:** 4 → 3

Configuration Actions

- Clone
- Create Draft
- Compare
- Restore
- Schedule
- Retire

Acceptance Condition:

Every material Wallet configuration change is version-controlled and historical configurations remain available for audit and investigation.

Screen 09 — Approval, Publication & Change Management

Purpose: Control how Wallet configuration moves safely from draft into production.

Publication Lifecycle

- **Draft**
- → **Validation**
- → **Simulation**
- → **Submitted**
- → **Business Approval**
- → **Security / Finance Approval where required**
- → **Approved**
- → **Scheduled**
- → **Published**
- → **Monitored**

Approval Matrix

Different changes can require different authorities.

For example:

- **Customer Display Change**
- → Product Owner
- **Consumption Rule**
- → Wallet Product Owner
- **Refund Rule**
- → Operations + Finance
- **Fraud Policy**
- → Security/Risk
- **Accounting Mapping**
- → Finance
- **API Security**
- → Security + Technology

Deployment Controls

Configure:

- Immediate publication
- Scheduled publication
- Tenant-specific rollout
- Venue-specific rollout
- Effective date
- Pilot rollout
- Emergency rollback

Change Impact Analysis

Before publication show:

- Wallets affected
- Customers affected
- Venues affected
- Credit types affected
- APIs affected
- Financial impact
- Security impact
- Integration impact

AI Capability:

AI identifies potentially conflicting changes across Wallet boards before publication.

Acceptance Condition:

No governed Wallet configuration enters production without satisfying its required validation, approval and publication workflow.

Screen 10 — Wallet Platform Health, Audit & Administration Center

Purpose: Provide the final master-administration screen for the entire TICVAI Wallet ecosystem.

This should become the **control tower for all ten Wallet boards**.

Platform Health

Display:

- **Wallet Engine**
Healthy

- **Wallet Ledger**
Healthy
- **Funding**
Healthy
- **Consumption Engine**
Healthy
- **Gift Cards**
Healthy
- **Wearables**
Healthy
- **Transfer Service**
Healthy
- **Fraud Engine**
Healthy
- **Finance Integration**
Healthy
- **API Gateway**
Healthy

Master Health KPIs

- Wallet Service Availability
- API Success Rate
- Average Authorization Time
- Failed Transactions
- Pending Events
- Reconciliation Exceptions
- Fraud Alerts
- Ledger Integrity
- Configuration Errors

Master Audit

Search across:

- Wallet creation
- Funding
- Spending
- Transfers
- Refunds
- Adjustments
- Expiration
- Credential actions
- Fraud actions
- Finance actions
- API actions
- Configuration changes
- Approvals
- Publications

Audit Record

Every record should contain:

- Audit ID
- Tenant
- Venue
- Wallet/customer where applicable
- Action

- Object
- User/system
- Old value
- New value
- Reason
- Approval
- Source
- Date/time
- Correlation/transaction ID

Administrative Controls

Authorized super administrators can:

- Run Health Check
- Run Configuration Validation
- Review Failed Services
- Review Integration Exceptions
- View Audit Trail
- Export Audit
- Compare Configuration
- Enter Maintenance Mode
- Suspend Integration
- Trigger controlled rollback

AI Wallet Health Assistant

Provide an AI-generated operational summary such as:

- **Wallet Platform Health: 96/100**

Three issues require attention.

1. Gift-card reconciliation has 24 unresolved exceptions totaling AED 18,450.
2. Venue 04 has a 6.2% increase in RFID authorization latency.
3. Finance API synchronization has 12 events pending longer than the configured SLA.

The administrator can drill directly into each issue.

Acceptance Condition:

Authorized administrators can assess end-to-end Wallet health, trace material actions across the platform and navigate directly to affected configuration, transactions, integrations or exceptions.

Board 10 — End-to-End Architecture

The final integration architecture should look like:

- Customer / Employee / Corporate Account
- ↓
- TICVAI Wallet Engine
- Wallet Foundation
 - → Funding
 - → Credit & Consumption
 - → Family / Corporate
 - → Gift Cards & Benefits
 - → Channels & Wearables
 - → Transfers & Refunds
 - → Fraud & Risk
 - → Finance & Liability
- ↓
- Wallet API & Event Layer
 - ↙ Ticketing
 - ↙ Membership
 - ↙ CRM
 - ↙ Pricing & Promotions
 - ↙ POS
 - ↙ B2C / Mobile App
 - ↙ F&B
 - ↙ Retail
 - ↙ Rental
 - ↙ Parking
 - ↙ Access Control
 - ↙ Payments
 - ↙ Finance
 - ↙ External Systems
- ↓
- Monitoring + Governance + Audit

Complete Wallet Module — 10 Boards / 100 Backend Screens

Board	Configuration Domain
01	Wallet Foundation & Wallet Types
02	Funding, Top-Up & Reload Management
03	Stored Value, Credits & Consumption Rules
04	Family, Parent–Child & Corporate Wallets
05	Gift Cards, Vouchers & Digital Benefits
06	Wallet Payments, Wearables & Channel Usage
07	Transfers, Refunds & Wallet Operations
08	Security, Fraud, Limits & AI Risk Controls
09	Finance, Liability, Reconciliation & Analytics
10	API, Integration, Governance & Wallet Administration

With Board 10, I would consider the **backend Wallet Configuration Module complete at 100 screens**. It covers the functional requirements while maintaining the important boundaries with Payments, Membership, Pricing, CRM and Finance instead of duplicating those modules.

The final design for Board 10 should visually feel like the **master control board** of the Wallet module: API health, integration topology, webhook monitoring, security permissions, synchronization, exceptions, version control, publication workflow and overall Wallet platform health all visible across the 10 screens.

